UNIT 1

1. **Artificial Intelligence primarily aims to:**

a) Replace hardware
b) Increase internet speed
c) Mimic human intelligence
d) Store large data

Answer: c) Mimic human intelligence

2. **The father of Artificial Intelligence is:**

a) Alan Turing
b) Geoffrey Hinton
c) John McCarthy
d) Elon Musk

Answer: c) John McCarthy

3. **AI integrates ideas from:**

a) Only computer science
b) Mathematics only
c) Multiple disciplines
d) Robotics only

Answer: c) Multiple disciplines

4. Which discipline contributes probability and statistics to AI?

a) Psychology
b) Mathematics
c) Philosophy
d) Linguistics

Answer: b) Mathematics

5. NLP stands for:

a) Neural Logic Programming
b) Natural Language Processing
c) Network Learning Process
d) Numerical Learning Program

Answer: b) Natural Language Processing

6. Machine Learning is a subset of:

a) Data Science
b) AI
c) Robotics
d) Statistics

Answer: b) AI

7. Deep Learning is a subset of:

a) AI
b) Robotics
c) Machine Learning
d) Data Mining

Answer: c) Machine Learning

8. Which AI approach focuses on rational thinking?

a) Act like humans
b) Think well
c) Act well
d) Behaviourist approach

Answer: b) Think well

9. ELIZA program is an example of:

a) Think rationally
b) Act like humans
c) Think like humans
d) Super AI

Answer: b) Act like humans

10. Narrow AI is also called:

a) Strong AI
b) General AI
c) Weak AI
d) Super AI

Answer: c) Weak AI

11. General AI currently:

a) Exists widely
b) Does not exist
c) Used in hospitals
d) Used in cars

Answer: b) Does not exist

12. Super AI is:

a) Available in labs
b) Used in gaming
c) Hypothetical
d) Weak AI

Answer: c) Hypothetical

13. Reactive machines:

a) Store memory
b) Learn from past
c) React to present only
d) Think independently

Answer: c) React to present only

14. IBM Deep Blue is:

a) General AI
b) Super AI
c) Reactive Machine
d) Theory of Mind AI

Answer: c) Reactive Machine

15. Self-driving cars are example of:

a) Reactive machines
b) Limited memory AI
c) Super AI
d) Self-aware AI

Answer: b) Limited memory AI

16. Theory of Mind AI focuses on:

a) Hardware
b) Human emotions
c) Data storage
d) Coding

Answer: b) Human emotions

17. Self-awareness AI:

a) Exists in robots
b) Exists in labs
c) Is hypothetical
d) Used in banking

Answer: c) Is hypothetical

18. Supervised learning uses:

a) Unlabeled data
b) Random data
c) Labeled data
d) Reinforcement signals

Answer: c) Labeled data

19. Spam detection is example of:

a) Unsupervised learning
b) Reinforcement learning
c) Supervised learning
d) Deep AI

Answer: c) Supervised learning

20. K-Means clustering belongs to:

a) Supervised learning
b) Reinforcement learning
c) Unsupervised learning
d) Deep learning

Answer: c) Unsupervised learning

21. Reinforcement learning works on:

a) Labels
b) Rewards and penalties
c) Manual coding
d) Static logic

Answer: b) Rewards and penalties

22. Q-learning is used in:

a) Supervised learning
b) Unsupervised learning
c) Reinforcement learning
d) Regression

Answer: c) Reinforcement learning

23. Deep Learning uses:

a) Decision trees
b) Neural networks
c) Sorting algorithms
d) Search trees

Answer: b) Neural networks

24. Deep Learning requires:

a) Small data
b) Medium data
c) Large datasets
d) No data

Answer: c) Large datasets

25. Neural networks are inspired by:

a) CPU architecture
b) Human brain
c) Hard disk
d) Internet

Answer: b) Human brain

26. Backpropagation is used to:

a) Increase speed
b) Adjust weights
c) Delete data
d) Encrypt files

Answer: b) Adjust weights

27. Data Science mainly focuses on:

a) Hardware
b) Extracting insights from data
c) Robotics
d) Automation only

Answer: b) Extracting insights from data

28. Deep Learning automatically:

a) Deletes features
b) Extracts features
c) Stores hardware
d) Encrypts data

Answer: b) Extracts features

29. One disadvantage of Deep Learning:

a) Needs small data
b) Requires low power
c) Hard to interpret
d) Easy to explain

Answer: c) Hard to interpret

30. AI in healthcare helps in:

a) Gaming
b) Diagnosis
c) Coding
d) Designing hardware

Answer: b) Diagnosis

31. AI in finance is used for:

a) Fraud detection
b) Cooking
c) Printing
d) Hardware design

Answer: a) Fraud detection

32. Recommendation systems use:

a) ML
b) OS
c) Hardware
d) BIOS

Answer: a) ML

33. Actor-Critic methods belong to:

a) Supervised learning
b) Reinforcement learning
c) Unsupervised learning
d) NLP

Answer: b) Reinforcement learning

34. GANs are used in:

a) Sorting
b) Image generation
c) Typing
d) Data storage

Answer: b) Image generation

35. Generative AI creates:

a) Only tables
b) Only code
c) New content
d) Hardware

Answer: c) New content

36. GPT is based on:

a) CNN
b) RNN
c) Transformer
d) KNN

Answer: c) Transformer

37. Self-attention is part of:

a) Neural trees
b) Transformers
c) Regression
d) Sorting

Answer: b) Transformers

38. Data poisoning attack affects:

a) Hardware
b) Training data
c) CPU
d) GPU

Answer: b) Training data

39. MFA stands for:

a) Multi-Function Access
b) Multi-Factor Authentication
c) Machine Feature Algorithm
d) Multiple Firewall Access

Answer: b) Multi-Factor Authentication

40. Overfitting means:

a) Model underperforms
b) Model memorizes training data
c) Model deletes data
d) Model encrypts

Answer: b) Model memorizes training data

41. IBM Watson is example of:

a) Super AI
b) Narrow AI
c) Self-aware AI
d) AGI

Answer: b) Narrow AI

42. Computer Vision deals with:

a) Text
b) Audio
c) Images and video
d) Numbers

Answer: c) Images and video

43. Logistic Regression is used in:

a) Supervised learning
b) Clustering
c) Reinforcement
d) Sorting

Answer: a) Supervised learning

44. PCA is used for:

a) Clustering
b) Dimensionality reduction
c) Encryption
d) Sorting

Answer: b) Dimensionality reduction

45. AI's goal is to:

a) Reduce RAM
b) Mimic intelligence
c) Increase battery
d) Format data

Answer: b) Mimic intelligence

46. Robotics uses AI for:

a) Manual work only
b) Intelligent automation
c) Storage
d) Deletion

Answer: b) Intelligent automation

47. AI-powered cybersecurity detects:

a) Hardware faults
b) Network anomalies
c) Printer issues
d) File size

Answer: b) Network anomalies

48. Deep Learning works best for:

a) Small tasks
b) Complex tasks
c) Manual coding
d) Basic math

Answer: b) Complex tasks

49. ML improves performance by:

a) Explicit programming
b) Learning from data
c) Deleting files
d) Resetting systems

Answer: b) Learning from data

50. ANI stands for:

a) Artificial Neural Intelligence
b) Artificial Narrow Intelligence
c) Advanced Network Intelligence
d) Autonomous Neural Interface

Answer: b) Artificial Narrow Intelligence

51. AGI stands for:

a) Artificial General Intelligence
b) Advanced Global Interface
c) Automated General Input
d) Artificial Graph Intelligence

Answer: a) Artificial General Intelligence

52. ASI refers to:

a) Artificial System Intelligence
b) Artificial Super Intelligence
c) Advanced Super Interface
d) Automated Smart Intelligence

Answer: b) Artificial Super Intelligence

53. Which type of AI can perform only a specific task?

a) General AI
b) Super AI
c) Narrow AI
d) Self-aware AI

Answer: c) Narrow AI

54. In supervised learning, output is:

a) Unknown
b) Labeled
c) Random
d) Encrypted

Answer: b) Labeled

55. Which learning method is used in customer segmentation?

a) Supervised learning
b) Unsupervised learning
c) Reinforcement learning
d) Deep learning only

Answer: b) Unsupervised learning

56. Reinforcement learning is commonly used in:

a) Email sorting
b) Game AI
c) Word processing
d) Spreadsheet design

Answer: b) Game AI

57. Decision Tree is an algorithm used in:

a) Supervised learning
b) Unsupervised learning
c) Reinforcement learning
d) Robotics only

Answer: a) Supervised learning

58. Deep learning models require:

a) Less computation
b) Moderate computation
c) High computational power
d) No computation

Answer: c) High computational power

59. Hidden layers are present in:

a) Basic algorithms
b) Neural networks
c) Search trees
d) Operating systems

Answer: b) Neural networks

60. Backpropagation is mainly used for:

a) Data cleaning
b) Weight adjustment
c) Encryption
d) Sorting

Answer: b) Weight adjustment

61. The input layer in neural network:

a) Produces output
b) Receives data
c) Deletes data
d) Encrypts data

Answer: b) Receives data

62. Overfitting occurs when:

a) Model performs well on new data
b) Model memorizes training data
c) Model deletes errors
d) Model underperforms

Answer: b) Model memorizes training data

63. One advantage of deep learning:

a) Requires small dataset
b) High interpretability
c) Automated feature extraction
d) Low computational need

Answer: c) Automated feature extraction

64. One disadvantage of deep learning:

a) Easy to interpret
b) Low accuracy
c) Requires large dataset
d) Needs no hardware

Answer: c) Requires large dataset

65. NLP helps machines to:

a) Build hardware
b) Understand human language
c) Repair software
d) Encrypt networks

Answer: b) Understand human language

66. Example of Generative AI:

a) K-Means
b) ChatGPT
c) Excel
d) Router

Answer: b) ChatGPT

67. GAN stands for:

a) General AI Network
b) Generative Adversarial Network
c) Global Artificial Node
d) Graphical AI Network

Answer: b) Generative Adversarial Network

68. Transformer architecture is mainly used in:

a) Hardware design
b) NLP tasks
c) Operating systems
d) Networking

Answer: b) NLP tasks

69. Self-attention mechanism helps to:

a) Delete tokens
b) Focus on relevant words
c) Encrypt data
d) Sort data

Answer: b) Focus on relevant words

70. Data Science overlaps with:

a) AI and ML
b) Hardware
c) BIOS
d) Firmware

Answer: a) AI and ML

71. Fraud detection mainly uses:

a) ML algorithms
b) Printers
c) Modems
d) RAM

Answer: a) ML algorithms

72. Robotics combined with AI leads to:

a) Manual control
b) Intelligent automation
c) Data deletion
d) Network failure

Answer: b) Intelligent automation

73. Image recognition is example of:

a) Computer Vision
b) Data entry
c) Encryption
d) Formatting

Answer: a) Computer Vision

74. Spam filtering is:

a) Unsupervised learning
b) Supervised learning
c) Reinforcement learning
d) Reactive AI

Answer: b) Supervised learning

75. Actor-Critic methods are used in:

a) Supervised learning
b) Unsupervised learning
c) Reinforcement learning
d) Regression only

Answer: c) Reinforcement learning

76. AI in manufacturing is used for:

a) Predictive maintenance
b) Formatting drives
c) Typing
d) Gaming

Answer: a) Predictive maintenance

77. Which AI type understands social interactions?

a) Limited memory
b) Reactive machine
c) Theory of Mind
d) Narrow AI

Answer: c) Theory of Mind

78. AI-powered anomaly detection is used in:

a) Gaming
b) Cybersecurity
c) Cooking
d) Painting

Answer: b) Cybersecurity

79. Deep learning performs feature extraction:

a) Manually
b) Automatically
c) Randomly
d) Sequentially

Answer: b) Automatically

80. Neural networks are useful for:

a) Linear problems only
b) Complex non-linear problems
c) Hardware repair
d) File deletion

Answer: b) Complex non-linear problems

81. Which ML type does not use labels?

a) Supervised
b) Reinforcement
c) Unsupervised
d) Deep

Answer: c) Unsupervised

82. Example of supervised algorithm:

a) K-Means
b) PCA
c) Decision Tree
d) Hierarchical clustering

Answer: c) Decision Tree

83. Example of unsupervised algorithm:

a) Logistic Regression
b) K-Means
c) SVM
d) Random Forest

Answer: b) K-Means

84. Reinforcement learning agent learns through:

a) Labels
b) Instructions
c) Rewards
d) Hardcoding

Answer: c) Rewards

85. AI in autonomous vehicles is used for:

a) Typing
b) Navigation
c) Formatting
d) Printing

Answer: b) Navigation

86. Data poisoning attack targets:

a) Hardware
b) Training data
c) CPU
d) RAM

Answer: b) Training data

87. Adversarial attack aims to:

a) Improve model
b) Deceive AI model
c) Train faster
d) Increase memory

Answer: b) Deceive AI model

88. MFA increases:

a) Storage
b) Security
c) Speed
d) RAM

Answer: b) Security

89. Philosophy contributes to AI in:

a) Ethics
b) Hardware
c) Coding
d) Formatting

Answer: a) Ethics

90. Psychology contributes to AI by studying:

a) CPU
b) Human behaviour
c) Hard disk
d) Networks

Answer: b) Human behaviour

91. Neuroscience inspired:

a) Firewalls
b) Neural networks
c) Modems
d) BIOS

Answer: b) Neural networks

92. Linguistics helps in:

a) NLP
b) Robotics
c) Hardware
d) Encryption

Answer: a) NLP

93. AI in retail is used for:

a) Hardware repair
b) Recommendation systems
c) Formatting
d) Coding

Answer: b) Recommendation systems

94. Deep Learning works best with:

a) Small data
b) Large data
c) No data
d) Static data

Answer: b) Large data

95. ML model generalizes to:

a) Old data only
b) New unseen data
c) No data
d) Corrupt data

Answer: b) New unseen data

96. GPU is required for:

a) Basic ML
b) Deep Learning training
c) Word processing
d) Printing

Answer: b) Deep Learning training

97. Generative AI mainly creates:

a) Hardware
b) Static code only
c) New content
d) Operating systems

Answer: c) New content

98. Foundation model is:

a) Small model
b) Large pre-trained model
c) Antivirus
d) Firewall

Answer: b) Large pre-trained model

99. Chatbots use:

a) NLP
b) BIOS
c) Hardware
d) Printer drivers

Answer: a) NLP

100. Generative AI is subset of:

a) AI
b) ML
c) Deep Learning
d) Data Science

Answer: c) Deep Learning

101.**Who is known as the father of Artificial Intelligence?**

(a) Alan Turing
(b) John McCarthy
(c) Andrew Ng
(d) Geoffrey Hinton

102.**Which discipline is NOT directly a part of AI core contributors?**

(a) Mathematics
(b) Psychology
(c) Astrology
(d) Linguistics

103.**What is the ultimate goal of Artificial Intelligence?**

(a) Only solve mathematical problems.

(b) Mimic human intelligence to solve complex tasks.
(c) Operate machines without human intervention.
(d) Build hardware devices.

104.**Which statement fits "Weak AI" or Narrow AI?**

(a) Has general human-level intelligence.
(b) Excels at multiple tasks simultaneously.
(c) Performs dedicated tasks within a limited scope.
(d) Can reason like a human.

105.**What differentiates Super AI from General AI?**

(a) Super AI has narrowly focused skills.
(b) General AI can exceed human intellectual performance.
(c) Super AI surpasses human intelligence in every domain.
(d) Super AI has no cognitive abilities.

106.**Reactive Machines in AI are characterized by ......**

(a) the ability to store past experiences.
(b) purely reacting to current inputs without memory.
(c) being self-aware.
(d) understanding human emotions.

107.**What is the major focus of Theory of Mind AI?**

(a) Data storage
(b) Understanding emotions and beliefs
(c) Performing calculations quickly
(d) Robot mechanics

108.**Which AI category is still hypothetical and involves consciousness?**

(a) Reactive Machines
(b) Limited Memory
(c) Self-Awareness AI
(d) Narrow AI

109.**AI helps in healthcare by ......**

(a) Diagnosing diseases and patient monitoring.
(b) Replacing doctors fully.
(c) Only recording patient data.
(d) Eliminating all human interaction.

110.**What role does linguistics play in AI?**

(a) Hardware design
(b) Natural Language Processing
(c) Robotics control
(d) Statistical analysis

111.**Machine Learning is best described as:**

(a) Programming specific instructions explicitly
(b) Training systems to learn from data
(c) Hardware manufacturing
(d) Software debugging

112.**Which is an example of supervised learning?**

(a) Clustering customer groups
(b) Email spam filtering with labeled emails
(c) Random data generation
(d) Exploring databases

113.**What algorithm is mainly used in supervised learning?**

(a) K-means clustering
(b) Logistic regression
(c) PCA
(d) Reinforcement learning

Answer: (b) Logistic regression
Explanation: Logistic regression is a common supervised learning algorithm.

114.**Unsupervised learning primarily involves:**

(a) Learning from labelled data.
(b) Discovering patterns from unlabelled data.
(c) Making decisions based on rewards.
(d) Pre-programmed rules.

Answer: (b) Discovering patterns from unlabelled data.
Explanation: Unsupervised learning works without labeled outputs.

115.**Reinforcement learning is based on ……**

(a) receiving rewards or penalties through trial and error.
(b) grouping data clusters.
(c) identifying keywords.
(d) using labeled data only.

Answer: (a) receiving rewards or penalties through trial and error.
Explanation: Reinforcement learning learns using feedback in the form of rewards or penalties.

116.**Which of the following is NOT an example of ML application?**

(a) Fraud detection.
(b) Spam detection.
(c) Chess playing with fixed rules only.
(d) Customer segmentation.

Answer: (c) Chess playing with fixed rules only.
Explanation: Fixed-rule systems do not involve learning, so they are not ML.

117.**Q-learning is a technique used in ……**

(a) Unsupervised Learning.
(b) Supervised Learning.
(c) Reinforcement Learning.
(d) Data Mining.

Answer: (c) Reinforcement Learning.
Explanation: Q-learning is a reinforcement learning algorithm.

118.**ML benefits retail mainly by ……**

(a) Robotics automation.
(b) Customer behaviour analysis and recommendations.
(c) Manufacturing.
(d) Network security.

Answer: (b) Customer behaviour analysis and recommendations.
Explanation: ML analyzes customer data to provide recommendations.

119.**Which technique is part of unsupervised learning?**

(a) Support Vector Machines.
(b) K-means clustering.
(c) Decision Trees.
(d) Reinforcement learning.

Answer: (b) K-means clustering.
Explanation: K-means is an unsupervised clustering algorithm.

120.**Which field overlaps with AI/ML but is focused on data-driven insights?**

(a) Physics.
(b) Data Science.
(c) Hardware Engineering.
(d) Linguistics.

Answer: (b) Data Science.
Explanation: Data Science focuses on extracting insights from data and overlaps with AI/ML.

121.**Deep Learning primarily uses ……**

(a) Decision Trees.
(b) Neural Networks with many layers.
(c) Statistical methods only.
(d) Simple linear regression.

Answer: (b) Neural Networks with many layers.
Explanation: Deep Learning is based on multi-layer neural networks.

122.**Deep Learning models excel at processing ……**

(a) Simple numerical data only.
(b) Complex unstructured data like images and speech.
(c) Only texts.
(d) Only numerical data.

Answer: (b) Complex unstructured data like images and speech.
Explanation: Deep Learning handles large and unstructured data efficiently.

123.**An advantage of Deep Learning is ......**

(a) Manual feature engineering.
(b) High accuracy in complex tasks.
(c) Low computational resources needed.
(d) Easy interpretability.

Answer: (b) High accuracy in complex tasks.
Explanation: Deep Learning achieves high accuracy in image, speech, and language tasks.

124.**A big challenge in Deep Learning is ......**

(a) Lack of data.
(b) Excessive interpretability.
(c) Simplicity of models.
(d) Universal understanding by humans.

Answer: (b) Excessive interpretability.
Explanation: Deep Learning models are difficult to interpret and act like a black box.

125.**GPUs are important in DL because they ......**

(a) Provide more storage.
(b) Accelerate complex computations efficiently.
(c) Reduce data requirements.
(d) Produce data labels.

Answer: (b) Accelerate complex computations efficiently.
Explanation: GPUs perform parallel processing required for deep learning.

126.**What is overfitting in Deep Learning?**

(a) Model performs well on new data only.
(b) Model fits training data too closely reducing generalization.
(c) Model training completed fast.
(d) Data normalization process.

Answer: (b) Model fits training data too closely reducing generalization.
Explanation: Overfitting occurs when a model learns training data too well and fails to perform on unseen data.

127.**Deep learning models improve with ……**

(a) More data and training time.
(b) Less computational power.
(c) Fixed data only.
(d) Manual intervention in feature selection.

Answer: (a) More data and training time.
Explanation: Deep learning models require large datasets and longer training to improve performance.

128.**Which layer in deep learning captures non-linear transformations?**

(a) Input Layer
(b) Hidden Layers
(c) Output Layer
(d) Data preprocessing

Answer: (b) Hidden Layers
Explanation: Hidden layers apply activation functions to capture complex non-linear patterns.

129.**An example of DL application in healthcare is ……**

(a) Algorithmic trading.
(b) Medical image analysis.
(c) Automated marketing.
(d) Cybersecurity threat detection.

Answer: (b) Medical image analysis.
Explanation: Deep learning is widely used in analyzing medical images such as X-rays and MRI scans.

130.**Which is a disadvantage of DL models?**

(a) They require minimal data.
(b) Difficult to interpret results (black-box).
(c) Easy to train on CPUs.
(d) Limited applicability.

Answer: (b) Difficult to interpret results (black-box).
Explanation: Deep learning models are complex and their internal decision-making is hard to explain.

131.**Neural networks are inspired by ……**

(a) Human brain neurons
(b) Computer circuits
(c) Statistical models
(d) Logistic regression

Answer: (a) Human brain neurons
Explanation: Neural networks are modeled after the structure and working of the human brain.

132.**The activation function in neural networks:**

(a) Summarizes inputs linearly.
(b) Normalizes weights.
(c) Introduces non-linearity.
(d) Connects layers.

Answer: (c) Introduces non-linearity.
Explanation: Activation functions enable neural networks to learn complex patterns.

133.**The learning process in neural networks mainly involves ……**

(a) manually adjusting weights.
(b) weight adjustment through backpropagation.
(c) clustering data.
(d) data labelling.

Answer: (b) weight adjustment through backpropagation.
Explanation: Backpropagation updates weights based on error to improve learning.

134.**What is weighted sum in neural networks?**

(a) Sum of all inputs without weights.
(b) Sum of inputs multiplied by weights.
(c) Average of output values.
(d) Total number of nodes.

Answer: (b) Sum of inputs multiplied by weights.
Explanation: Each input is multiplied by a weight and summed before activation.

135.**Neural networks are widely used for ……**

(a) Pattern recognition and classification.
(b) Operating systems.
(c) Hardware design.
(d) Network set-up.

Answer: (a) Pattern recognition and classification.
Explanation: Neural networks excel at recognizing patterns in data.

136.**What is the output layer's job?**

(a) Process input data.
(b) Adjust weights during training.
(c) Generate final model prediction.
(d) Store training data.

Answer: (c) Generate final model prediction.
Explanation: The output layer provides the final result of the neural network.

137.**Why are neural networks sometimes called "black box" systems?**

(a) Because they are simple to understand.
(b) Their internal decision processes are complex and hard to interpret.
(c) Because they use black colored hardware.
(d) They do not produce outputs.

Answer: (b) Their internal decision processes are complex and hard to interpret.
Explanation: The hidden layers make neural networks difficult to interpret.

138.**Which method helps neural networks learn from error?**

(a) Forward propagation.
(b) Backpropagation.
(c) Data preprocessing.
(d) Parameter tuning.

Answer: (b) Backpropagation.
Explanation: Backpropagation adjusts weights based on error feedback.

139.**Layers between input and output layers are called ……**

(a) Data layers.
(b) Hidden layers.
(c) Input layers.
(d) Output layers.

Answer: (b) Hidden layers.
Explanation: Hidden layers process data between input and output.

140.**Neural networks can adapt and recognize patterns better than ……**

(a) Rule-based systems.
(b) Linear models.
(c) Traditional programming.
(d) All of the above.

Answer: (d) All of the above.
Explanation: Neural networks outperform traditional methods in complex pattern recognition.

141. **Generative AI is designed to ……**

(a) Analyze but not create content
(b) Create new content from learned patterns
(c) Only classify data
(d) Run hardware operations

Answer: (b) Create new content from learned patterns
Explanation: Generative AI produces new text, images, audio, and video.

142. **Which data format is NOT used to train Generative AI?**

(a) Text
(b) Images
(c) 3D signals
(d) Only numeric tables

Answer: (d) Only numeric tables
Explanation: Generative AI uses diverse data formats, not just numeric tables.

143. **The "Foundation Model" in Generative AI is ……**

(a) A small lightweight model.
(b) A large neural network trained on massive diverse data.
(c) A simple decision tree.
(d) A rule-based system.

Answer: (b) A large neural network trained on massive diverse data.
Explanation: Foundation models learn general knowledge from large datasets.

144. **The self-attention mechanism in Transformers helps ……**

(a) Focus on all parts of the input sequence for context.
(b) Filter irrelevant data before training.
(c) Generate random outputs.

(d) Only translate languages.

Answer: (a) Focus on all parts of the input sequence for context.
Explanation: Self-attention relates every word to every other word.

145.**Multi-head attention means ……**

(a) Running several attention layers in parallel.
(b) One attention head per sequence.
(c) Removing attention heads.
(d) Simple output generation.

Answer: (a) Running several attention layers in parallel.
Explanation: Multiple heads capture different relationships in data.

146.**Positional encoding in Transformers is used because ……**

(a) Transformers have no inherent sequence processing.
(b) To encode data size.
(c) To speed-up training.
(d) To eliminate tokens.

Answer: (a) Transformers have no inherent sequence processing.
Explanation: Positional encoding provides word order information.

147.**Which is NOT a type of Generative AI?**

(a) Text generation
(b) Image generation
(c) Database management
(d) Music generation

Answer: (c) Database management
Explanation: Database management is not content generation.

148.**An example of text generation model is ……**

(a) DALL·E
(b) GPT
(c) MidJourney
(d) Jukebox

Answer: (b) GPT
Explanation: GPT generates human-like text.

149.**One application of Generative AI in finance is ……**

(a) Fraud detection.
(b) Automated report generation.
(c) Stock trading without AI.
(d) Network security hardware.

Answer: (b) Automated report generation.
Explanation: Generative AI creates financial reports automatically.

150.**Transformer models use which processing architecture?**

(a) Encoder-decoder
(b) Recurrent networks
(c) Feedforward only
(d) Clustering layers

Answer: (a) Encoder-decoder
Explanation: Transformers use encoder-decoder architecture to process and generate sequences.

UNIT 2- Internet of Things

1. Which of the following is a key characteristic of 5G that enables real-time applications like autonomous driving?

a) High power consumption
b) Ultra-low latency
c) Limited device connectivity
d) Low data rates

Answer:
b) Ultra-low latency

Explanation:
Ultra-low latency is a defining feature of 5G that allows data to be transmitted with minimal delay, which is essential for real-time applications such as autonomous vehicles and remote surgery.

2. What is the typical peak data rate supported by 5G networks?

a) 1 Gbps10 Gbps
b) 20 Gbps
c) 100 Mbps

Answer:
c) 20 Gbps

Explanation:
5G networks are designed to support very high peak data rates, theoretically reaching up to 20 Gbps under ideal conditions.

3. Which 5G feature allows it to support up to 1 million devices per square kilometre?

a) Network slicing
b) Massive Machine-Type Communications (mMTC)
c) Ultra-Reliable Low-Latency Communications (URLLC)
d) Beamforming

Answer:
b) Massive Machine-Type Communications (mMTC)

Explanation:
mMTC enables connectivity for a very large number of IoT devices in a small area, making it suitable for smart cities and large-scale IoT deployments.

4. Which application area of 5G is most associated with real-time remote surgeries?

a) Smart agriculture
b) Healthcare
c) Smart homes
d) Logistics

Answer:
b) Healthcare

Explanation:
5G supports remote surgeries through ultra-low latency and high reliability, which are critical in medical procedures.

5. What is the purpose of network slicing in 5G technology?

a) To increase power consumption
b) To create multiple virtual networks on the same infrastructure
c) To reduce network coverage
d) To limit device connectivity

Answer:
b) To create multiple virtual networks on the same infrastructure
Explanation:
Network slicing allows multiple virtual networks to be created over a single physical network, each optimized for different services such as IoT, healthcare, or video streaming.

6. Which of the following is an application of 5G in smart cities?

a) Remote gaming
b) Traffic management with real-time sensor data
c) Virtual reality training for surgeons
d) Precision farming

Answer:
b) Traffic management with real-time sensor data

Explanation:
5G enables real-time data collection and processing from sensors, helping in efficient traffic management in smart cities.

7. Which 5G characteristic improves energy efficiency for IoT devices?

a) High latency

b) Optimized power consumption
c) Limited bandwidth
d) Low reliability

Answer:
b) Optimized power consumption

Explanation:
5G optimizes power usage, allowing IoT devices to operate efficiently for longer durations with lower energy consumption.

8. IoT stands for ………

a) Internet of Technology
b) Intranet of Things
c) Internet of Things
d) Information of Things

Answer:
c) Internet of Things

Explanation:
IoT stands for Internet of Things, which refers to a network of interconnected physical objects that collect and exchange data over the internet.

9. What is the Internet of Things (IoT)?

a) A technology used only for smartphones
b) A network of interconnected physical devices that exchange data
c) A type of computer virus
d) A communication protocol for websites

Answer:
b) A network of interconnected physical devices that exchange data

Explanation:
IoT connects physical devices embedded with sensors and connectivity, enabling them to communicate and share data.

10. Which of the following is a key characteristic of IoT?

a) Manual operation
b) Isolation of devices
c) Interconnectivity

d) Lack of automation

Answer:
c) Interconnectivity

Explanation:
Interconnectivity allows IoT devices to communicate with each other and with cloud systems over networks.

11. Which IoT feature enables devices to operate without human involvement?

a) Automation
b) Virtualization
c) Encryption
d) Translation

Answer:
a) Automation

Explanation:
Automation allows IoT devices to perform actions automatically based on predefined conditions and sensor inputs.

12. 'Things' in IoT refers to ....

a) Only smartphones
b) Any physical object with a sensor and connectivity
c) Website links
d) Only computers

Answer:Any physical object with a sensor and connectivity

Explanation:
In IoT, "things" are physical objects equipped with sensors, software, and network connectivity.

13. Which communication technology is widely used in IoT?

a) HDMI
b) USB
c) Wi-Fi
d) VGA

Answer:
c) Wi-Fi

Explanation:
Wi-Fi is commonly used in IoT for wireless communication between devices and the internet.

14. Which of the following is an application of IoT?

a) Social media chatting
b) Smart home automation
c) Playing offline games
d) File compression
Answer:
b) Smart home automation

Explanation:
Smart home automation uses IoT devices to control lighting, security, appliances, and energy systems automatically.

15. IoT allows devices to collect and share data using ....
a) Sensors
b) CDs
c) Printers
d) Speakers
Answer:
a) Sensors

Explanation:
Sensors are the primary components that collect environmental data in IoT systems.

16. Which is an advantage of IoT?
a) Reduced automation
b) Increased human effort
c) Real-time monitoring
d) Less data availability
Answer:
c) Real-time monitoring

Explanation:
IoT enables continuous and real-time monitoring of systems, improving efficiency and decision-making.

17. Which of the following is a limitation of IoT?
a) High scalability
b) Security and privacy issues
c) Increased efficiency
d) Better data collection

Answer:
b) Security and privacy issues

Explanation:
IoT systems are vulnerable to cyberattacks and data breaches, making security a major concern.

18. Which characteristic of IoT enables devices to work together?

a) Fragmentation
b) Interoperability
c) Decoupling
d) Sampling

Answer:
b) Interoperability

Explanation:
Interoperability allows different IoT devices, platforms, and systems to communicate and function together smoothly.

19. Smart watches, fitness trackers, and medical sensors fall under which IoT application area?

a) Industrial automation
b) Healthcare
c) Transport
d) Agriculture

Answer:
b) Healthcare

Explanation:
Wearable devices and medical sensors are used to monitor health parameters, making them part of healthcare IoT applications.

20. Which of the following is NOT an IoT advantage?

a) Improved monitoring
b) Better decision-making
c) Increased cyber risks
d) Automation

Answer:
c) Increased cyber risks

Explanation:
Increased cyber risks are a disadvantage of IoT due to security vulnerabilities, not an advantage.

21. Cloud computing helps IoT by ………

a) Storing and processing large amounts of data
b) Creating viruses
c) Blocking device communication
d) Removing internet access
Answer:
a) Storing and processing large amounts of data

Explanation:
Cloud computing provides storage, processing, and analytics capabilities required to handle massive IoT data.

22. Which component is essential in IoT devices?

a) Sensors and actuators
b) Chalk and board
c) Pen drive
d) Hard disk
Answer:
a) Sensors and actuators

Explanation:
Sensors collect data and actuators perform actions, making them essential components of IoT systems.

23. Which IoT application is widely used in agriculture?

a) Weather-based irrigation systems
b) Online shopping portals
c) Music streaming
d) Video editing
Answer:
a) Weather-based irrigation systems

Explanation:
IoT-based irrigation systems use sensors to monitor weather and soil conditions to optimize water usage.

24. Physical design of IoT mainly focuses on ………

a) Data flow
b) Hardware components
c) Cloud services
d) Software standards

Answer:
b) Hardware components

Explanation:
Physical design deals with sensors, actuators, devices, and hardware used in IoT systems.

25. Which of the following is part of the physical design of IoT?

a) IoT protocols
b) Sensors and actuators
c) Cloud databases
d) Data analytics

Answer:
b) Sensors and actuators

Explanation:
Sensors and actuators are physical components directly involved in interacting with the environment.

26. Logical design of IoT deals with ………

a) Hardware wiring
b) Functional blocks and data flows
c) Power supply
d) Battery types
Answer:
b) Functional blocks and data flows

Explanation:
Logical design focuses on how different functional blocks interact and how data flows through the system.

27. Which is NOT a physical component of IoT?

a) Sensors
b) RFID tags
c) Application layer
d) Actuators

Answer:
c) Application layer

Explanation:
The application layer belongs to the logical/software architecture, not the physical components.

28. An actuator in IoT is used for ………

a) Receiving data
b) Storing data
c) Performing actions
d) Testing networks

Answer:
c) Performing actions

Explanation:
Actuators convert electrical signals into physical actions such as switching on motors or lights.

29. Which device converts physical signals into electrical signals?

a) Actuator
b) Sensor
c) Router
d) Microcontroller

Answer:
b) Sensor

Explanation:
Sensors detect physical quantities like temperature or pressure and convert them into electrical signals.

30. Which of the following is part of logical design?

a) RFID
b) MQTT protocol
c) ESP32 board
d) Temperature sensor
Answer:
b) MQTT protocol

Explanation:
Logical design of IoT includes communication models and communication APIs such as MQTT. RFID, ESP32, and temperature sensors belong to physical design.

31. IoT architecture commonly uses how many layers?

a) 1
b) 3
c) 5
d) 7
Answer:
b) 3

Explanation:
The commonly used IoT architecture consists of three layers: Perception layer, Network layer, and Application layer.

32. The perception layer in IoT architecture handles ………

a) Data processing
b) Sensing and data collection
c) Application services
d) Communication security
Answer:
b) Sensing and data collection

Explanation:
The perception layer consists of sensors and actuators that sense physical parameters and collect data from the environment.

33. The network layer in IoT architecture is responsible for ………

a) Storing data only
b) Processing raw data
c) Transmitting data
d) Displaying output
Answer:
c) Transmitting data

Explanation:
The network layer transfers data collected by sensors to the cloud or application layer using communication technologies.

34. Application layer in IoT handles ………

a) Signal amplification
b) User-side services
c) Device manufacturing
d) Power management

Answer:
b) User-side services

Explanation:
The application layer provides services to users such as dashboards, monitoring systems, and control applications.

35. Which of these is a communication technology used in physical design?

a) Wi-Fi
b) Excel
c) Twitter
d) HTML
Answer:
a) Wi-Fi

Explanation:
Wi-Fi is a communication technology used in the physical design of IoT to enable device connectivity.

36. IoT logical design uses functional blocks such as ....

a) Power blocks
b) Communication blocks
c) Display blocks
d) Motor blocks
Answer:
b) Communication blocks

Explanation:
Logical design includes functional blocks like device, communication, services, management, security, and application blocks.

37. Which of the following is part of the IoT functional block?

a) Security
b) Shoes
c) Bulb
d) Wire

Answer:
a) Security

Explanation:
Security is an IoT functional block responsible for authentication, authorization, and data protection.

38. Physical design includes which component?

a) Cloud API
b) Sensors
c) SQL database
d) Data warehouse

Answer:
b) Sensors

Explanation:
Sensors are physical components used to collect data from the environment, hence part of physical design.

39. The main job of the IoT communication block is ....

a) To cook food
b) To transfer data between devices
c) To store raw data
d) To protect files

Answer:
b) To transfer data between devices

Explanation:
The communication block manages data transmission between IoT devices, gateways, servers, and cloud platforms.

40. In IoT architecture, which layer handles data routing?

a) Application
b) Network
c) Perception
d) Storage
Answer:
b) Network

Explanation:
The network layer handles routing and transmission of data across networks in an IoT system.

41. Logical design describes ………

a) How the device physically looks
b) How IoT functions conceptually
c) Voltage requirements
d) Mechanical shape
Answer:
b) How IoT functions conceptually

Explanation:
Logical design focuses on the conceptual working of IoT systems, including functional blocks, data flow, and communication models, not physical appearance.

42. Which of these is a sensing component?

a) LED
b) Temperature sensor
c) Speaker
d) Switch
Answer:
b) Temperature sensor

Explanation:
A temperature sensor detects physical temperature and converts it into an electrical signal, making it a sensing component.

43. The 3-layer IoT architecture is also called ………

a) Simple IoT architecture
b) Basic reference model
c) Open IoT structure
d) Smart architecture
Answer:
b) Basic reference model

Explanation:
The 3-layer IoT architecture (Perception, Network, Application) is commonly referred to as the basic reference model.

44. RFID tags are used in IoT for ………

a) Audio output
b) Object identification
c) Multimedia sharing
d) Gaming
Answer:
b) Object identification

Explanation:
RFID tags are mainly used to uniquely identify objects in IoT systems.

45. Micro-controllers like Arduino belong to ………

a) Logical design
b) Physical design
c) Database architecture
d) Application layer
Answer:
b) Physical design

Explanation:
Microcontrollers are hardware components and hence part of the physical design of IoT.

46. Logical design defines ………

a) IoT network placement
b) IoT communication protocols
c) IoT battery size
d) Motor speed
Answer:
b) IoT communication protocols

Explanation:
Logical design includes communication models and protocols that define how IoT components interact.

47. The physical design determines ………

a) Cloud storage type
b) Actual IoT device components
c) Software algorithms
d) User interface

Answer:
b) Actual IoT device components

Explanation:
Physical design deals with real hardware such as sensors, actuators, microcontrollers, and communication modules.

48. Which architecture layer is closest to the IoT device?

a) Application
b) Network
c) Perception
d) Middleware

Answer:
c) Perception

Explanation:
The perception layer directly interacts with physical devices through sensors and actuators.

49. In IoT, "Things" refer to ………

a) Only mobile phones
b) Physical devices with sensors
c) Software applications
d) Audio devices only
Answer:
b) Physical devices with sensors

Explanation:
"Things" in IoT are physical objects embedded with sensors and connectivity.

50. Which is NOT included in logical design?

a) Data flow
b) IoT Protocols
c) Hardware sensors
d) Functional blocks
Answer:
c) Hardware sensors

Explanation:
Hardware sensors belong to physical design, not logical design.

51. Fog computing belongs to which layer?

a) Perception
b) Application
c) Network / Middleware
d) Hardware
Answer:
c) Network / Middleware

Explanation:
Fog computing operates between the cloud and IoT devices, providing computation, storage, and networking services closer to the data source. Hence, it belongs to the Network/ Middleware layer.

52. Which protocol is used in IoT logical design for messaging?

a) MQTT
b) HDMI
c) VGA
d) FTP
Answer:
a) MQTT

Explanation:
MQTT is a lightweight messaging protocol used in IoT logical design for efficient data communication between devices.

53. A common physical IoT component for connectivity is ....

a) ESP8266 module
b) Adobe Reader
c) PowerPoint
d) Facebook app
Answer:
a) ESP8266 module

Explanation:
ESP8266 is a Wi-Fi module widely used in IoT devices to provide wireless connectivity, making it a common physical IoT component.

54. IoT architecture focuses on ....

a) Device modelling
b) Layered communication
c) Movie editing
d) Painting

Answer:
b) Layered communication

Explanation:
IoT architecture is designed using layers such as perception, network, and application to manage communication and data flow efficiently.

55. Which layer ensures user-specific IoT services?

a) Network
b) Application
c) Perception
d) Transport

Answer:
b) Application

Explanation:
The application layer provides user-specific services like dashboards, alerts, and control interfaces.

56. The term "IoT functional view" refers to ....

a) Logical design
b) Clothing design
c) Hardware manufacturing
d) Android coding
Answer:
a) Logical design

Explanation:
The functional view of IoT explains how components work conceptually, which is part of logical design.

57. In logical design, data processing is part of ....

a) Functional block
b) Battery block
c) Physical tool
d) Mechanical block

Answer:
a) Functional block

Explanation:
Logical design uses functional blocks such as sensing, processing, communication, and control.

58. Which of the following best describes IoT architecture?

a) A layered structure
b) A flat structure
c) A random structure
d) A circular structure

Answer:
a) A layered structure

Explanation:
IoT architecture is organized into layers to separate responsibilities and improve scalability.

59. An example of a physical design communication device is ………

a) LDR sensor
b) Bluetooth module
c) SQL server
d) Cloud analytics

Answer:
b) Bluetooth module

Explanation:
Bluetooth modules are physical hardware components used for wireless communication in IoT.

60. Which layer includes device addressing?

a) Application
b) Network
c) Perception
d) Security layer
Answer:
b) Network

Explanation:
The network layer handles device addressing, routing, and data transmission.

61. Sensors and actuators are part of which layer?

a) Network
b) Perception
c) Application
d) Processing

Answer:
b) Perception

Explanation:
The perception layer includes sensors and actuators that interact with the physical environment.

62. Which is a key function of IoT architecture?

a) Organizing data flow
b) Video editing
c) Graphic design
d) Gaming support
Answer:
a) Organizing data flow

Explanation:
IoT architecture organizes how data is collected, transmitted, processed, and used.

63. A gateway in IoT is mainly used for ………

a) Cooking
b) Bridging devices with the internet
c) Taking photos
d) Increasing sound
Answer:
b) Bridging devices with the internet

Explanation:
IoT gateways connect local devices and sensors to the internet or cloud.

64. Logical design focuses on ………

a) Device structure
b) Conceptual technical representation
c) Physical body
d) Mechanical shape
Answer:
b) Conceptual technical representation

Explanation:
Logical design explains how IoT components function and interact conceptually.

65. Which of the following is a physical IoT device?

a) Cloud software
b) Wi-Fi router
c) Big data analytics
d) HTML code

Answer:
b) Wi-Fi router

Explanation:
A Wi-Fi router is a physical hardware component used to connect IoT devices to networks.

66. The architecture layer that makes decisions for user is ………

a) Perception
b) Application
c) Network
d) Physical

Answer:
b) Application

Explanation:
The application layer processes data and makes decisions such as alerts, control actions, and user-level responses.

67. Middleware IoT architecture often lies between ………

a) Application and hardware
b) Network and perception
c) User and government
d) Internet and email
Answer:
a) Application and hardware

Explanation:
Middleware acts as a bridge between physical devices (hardware) and applications, enabling communication and data processing.

68. IoT protocols mainly belong to ………

a) Logical design
b) Physical design
c) Robotics
d) Mathematics

Answer:
a) Logical design

Explanation:
Protocols define communication rules and data exchange, which are part of the logical design of IoT.

69. The physical design of IoT includes which of these?

a) Wi-Fi module
b) XML
c) SQL
d) Website
Answer:
a) Wi-Fi module

Explanation:
Physical design includes hardware components such as sensors, actuators, and communication modules like Wi-Fi.

70. The operation of sensors in IoT is part of ....

a) Logical design
b) Physical design
c) Application design
d) Cloud design
Answer:
b) Physical design

Explanation:
Sensors are physical devices, so their operation is included in the physical design of IoT.

71. A sensor is a device that ....

a) Performs physical action
b) Converts physical quantity into electrical signals
c) Stores data
d) Only transmits data

Answer:
b) Converts physical quantity into electrical signals

Explanation:
Sensors detect physical parameters like temperature or pressure and convert them into electrical signals.

72. Which of the following is an example of an actuator?

a) Temperature sensor
b) Soil moisture sensor
c) DC motor
d) LDR sensor

Answer:
c) DC motor

Explanation:
Actuators perform actions. A DC motor converts electrical energy into mechanical motion.

73. A PIR sensor is used to detect ………

a) Light
b) Pressure
c) Motion
d) Sound
Answer:
c) Motion

Explanation:
PIR (Passive Infrared) sensors detect motion by sensing changes in infrared radiation.

74. Which sensor is used for measuring temperature?

a) LDR
b) Ultrasonic sensor
c) LM35
d) IR sensor
Answer:
c) LM35

Explanation:
LM35 is a commonly used temperature sensor that provides output proportional to temperature.

75. Which of the following detects light intensity?

a) DHT11
b) LDR
c) Load cell
d) MQ-2

Answer:
b) LDR

Explanation:
LDR (Light Dependent Resistor) changes resistance based on light intensity.

76. The MQ-2 sensor is used to detect ....

a) Motion
b) Gas
c) Water level
d) Pressure

Answer:
b) Gas

Explanation:
MQ-2 is a gas sensor used for detecting gases like LPG, smoke, and methane.

77. Which actuator converts electrical energy into rotational motion?

a) Relay
b) DC motor
c) LDR
d) Thermistor
Answer:
b) DC motor

Explanation:
A DC motor converts electrical energy into rotational mechanical motion.

78. A servo motor is an actuator used for ....

a) Temperature sensing
b) Precise angular movement
c) Sound detection
d) Light measurement

Answer:
b) Precise angular movement

Explanation:
Servo motors are used where accurate angle control is required, such as in robotics.

79. Which sensor is used for distance measurement in IoT?

a) Ultrasonic sensor
b) Thermistor
c) IR flame sensor
d) MQ-135

Answer:
a) Ultrasonic sensor

Explanation:
Ultrasonic sensors measure distance by sending sound waves and calculating the time taken for the echo to return.

80. Which sensor is commonly used in smart homes for humidity and temperature?

a) LDR
b) DHT11
c) Load cell
d) Motion sensor
Answer:
b) DHT11

Explanation:
DHT11 is a combined temperature and humidity sensor widely used in smart home and IoT projects.

81. Load cells are used to measure ………

a) Light
b) Gas
c) Weight / force
d) Altitude
Answer:
c) Weight / force

Explanation:
Load cells convert applied force or weight into electrical signals.

82. Which of these is NOT a sensor?

a) Gyroscope
b) Accelerometer
c) Relay
d) Thermistor

Answer:
c) Relay

Explanation:
A relay is an actuator/switch, not a sensor. Sensors only detect physical parameters.

83. An actuator that allows switching of high-voltage devices is ………

a) Servo motor
b) Relay
c) LDR
d) IR sensor

Answer:
b) Relay

Explanation:
Relays are used to control high-voltage or high-current devices using low-power signals.

84. A thermistor is used for ………

a) Temperature measurement
b) Sound detection
c) Distance measurement
d) Vibration detection
Answer:
a) Temperature measurement

Explanation:
Thermistors change resistance with temperature and are used for temperature sensing.

85. Which sensor is used in smartphones for screen rotation?

a) Gyroscope
b) Gas sensor
c) PIR sensor
d) LDR

Answer:
a) Gyroscope

Explanation:
Gyroscopes detect orientation and angular movement, enabling screen rotation.

86. Accelerometers measure ………

a) Temperature
b) Gas concentration
c) Acceleration / tilt
d) Noise level

Answer:
c) Acceleration / tilt

Explanation:
Accelerometers measure acceleration and tilt, useful in motion detection.

87. A buzzer in IoT acts as a ………

a) Sensor
b) Actuator
c) Controller
d) Storage device
Answer:
b) Actuator

Explanation:
A buzzer performs an action (sound output), so it is an actuator.

88. Which sensor is used in smoke detection?

a) MQ-2
b) BMP180
c) LDR
d) PIR
Answer:
a) MQ-2

Explanation:
MQ-2 detects smoke and combustible gases, commonly used in fire alarm systems.

89. Which sensor is used to detect fire?

a) LDR
b) Flame sensor
c) DHT11
d) Ultrasonic

Answer:
b) Flame sensor

Explanation:
Flame sensors detect infrared radiation emitted by fire.

90. A stepper motor is used when ………

a) Rotation in steps is required
b) Gas detection is needed
c) Temperature must be sensed
d) Light measurement is needed
Answer:
a) Rotation in steps is required

Explanation:
Stepper motors rotate in discrete steps, allowing precise position control.

91. Which sensor detects sound levels?

a) BMP180
b) Sound sensor
c) Ultrasonic
d) Gyroscope
Answer:
b) Sound sensor

Explanation:
Sound sensors detect sound intensity or noise levels.

92. The BMP180 sensor is used for ………

a) Gas detection
b) Altitude and pressure
c) Vibration measurement
d) Light sensing

Answer:
b) Altitude and pressure

Explanation:
BMP180 is a barometric pressure sensor used for altitude estimation.

93. A water level sensor is used to ………

a) Monitor humidity
b) Detect water depth or presence
c) Measure gas
d) Control motor speed

Answer:
b) Detect water depth or presence

Explanation:
Water level sensors detect the presence or level of water in tanks or reservoirs.

94. Which of the following is a digital sensor?

a) LDR
b) Thermistor
c) DHT22
d) Load cell
Answer:
c) DHT22

Explanation:
DHT22 provides digital output for temperature and humidity readings.

95. The IR sensor mainly detects ………

a) Infrared radiation and object presence
b) Temperature
c) Water flow
d) Gas leakage
Answer:
a) Infrared radiation and object presence

Explanation:
IR sensors detect infrared radiation reflected from objects.

96. A relay can be classified as ………

a) Input sensor
b) Output actuator

c) Storage device
d) Power supply

Answer:
b) Output actuator

Explanation:
Relays perform switching actions, so they are actuators.

97. Which sensor is suitable for soil moisture measurement?

a) PIR
b) Capacitive soil moisture sensor
c) MQ-135
d) Ultrasonic

Answer:
b) Capacitive soil moisture sensor

Explanation:
Capacitive soil moisture sensors measure water content in soil.

98. A vibration sensor is used for ………

a) Detecting movement in machines
b) Measuring humidity
c) Reading light intensity
d) Checking pressure

Answer:
a) Detecting movement in machines

Explanation:
Vibration sensors detect abnormal vibrations in machinery.

99. A photodiode is primarily used for ………

a) Gas detection
b) Light sensing
c) Sound detection
d) Temperature measurement
Answer:
b) Light sensing

Explanation:
Photodiodes convert light into electrical current.

100. Which of these is an example of an actuator used in home automation?

a) Ultrasonic sensor
b) Gas sensor
c) Electric door lock
d) DHT11

Answer:
c) Electric door lock

Explanation:
Electric door locks perform actions (locking/unlocking), so they are actuators.

101. A flow sensor is used to measure ………

a) Gas
b) Fluid movement
c) Sound
d) Pressure

Answer:
b) Fluid movement

Explanation:
Flow sensors measure the rate of liquid or gas flow in pipelines.

102. The role of an actuator in IoT is to ………

a) Process data
b) Sense environmental changes
c) Perform an action based on input
d) Store cloud information

Answer:
c) Perform an action based on input

Explanation:
An actuator converts electrical signals into physical actions such as motion, switching, or control.

103. Which sensor is commonly used in IoT-based weather monitoring systems?

a) PIR sensor
b) DHT22
c) Ultrasonic sensor
d) Relay

Answer:
b) DHT22

Explanation:
DHT22 measures temperature and humidity, which are key weather parameters.

104. Which actuator is used to control water flow in smart irrigation systems?

a) Servo motor
b) Solenoid valve
c) LDR
d) Gyroscope
Answer:
b) Solenoid valve

Explanation:
Solenoid valves open or close water flow electronically in irrigation systems.

105. A Hall Effect sensor is used to measure ………

a) Temperature
b) Magnetic field
c) Humidity
d) Sound
Answer:
b) Magnetic field

Explanation:
Hall Effect sensors detect magnetic fields and are used in speed and position sensing.

106. 5G networks are mainly designed to support ………

a) Only voice calls
b) Low-speed IoT devices
c) High-speed, low-latency communication
d) Cable TV signals

Answer:
c) High-speed, low-latency communication

Explanation:
5G provides ultra-low latency and very high data rates for advanced applications.

107. Which of the following is a key characteristic of 5G?

a) High latency
b) Low bandwidth
c) Ultra-low latency
d) Limited connectivity

Answer:
c) Ultra-low latency

Explanation:
Ultra-low latency enables real-time applications like autonomous driving and remote surgery.

108. 5G supports IoT using which feature?

a) Massive machine-type communication (mMTC)
b) Dial-up networking
c) USB connections
d) Satellite TV signals
Answer:
a) Massive machine-type communication (mMTC)

Explanation:
mMTC allows millions of IoT devices to connect simultaneously.

109. One important application area of 5G-IoT is ………

a) Offline gaming
b) Smart cities
c) Paper printing
d) Manual farming
Answer:
b) Smart cities

Explanation:
5G enables smart traffic, energy management, and public safety in smart cities.

110. 5G can support up to ………

a) Dozens of devices
b) 100 devices

c) Millions of devices per square kilometer
d) Only one device per household

Answer:
c) Millions of devices per square kilometer

Explanation:
5G is designed for massive device connectivity required by IoT ecosystems.

111. Which frequency band is commonly used in 5G?

a) Low-frequency AM band
b) 700 MHz – 52 GHz
c) 10–20 kHz
d) 1–5 kHz

Answer:
b) 700 MHz – 52 GHz

Explanation:
5G operates across sub-6 GHz and millimeter-wave frequency bands.

112. Which characteristic of 5G enables real-time IoT operations?

a) High latency
b) Low latency
c) Weak signal strength
d) Slow transmission

Answer:
b) Low latency

Explanation:
Low latency ensures instant data transmission for time-critical IoT applications.

113. An example of extreme real-time communication using 5G is ………

a) File transfer
b) Remote surgery
c) Email sending
d) Social media browsing

Answer:
b) Remote surgery

Explanation:
Remote surgery requires ultra-low latency and high reliability provided by 5G.

114. 5G IoT applications in transportation include ………

a) Self-driving cars
b) Typewriters
c) Postal mail
d) Bicycle sensors only

Answer:
a) Self-driving cars

Explanation:
Autonomous vehicles rely on fast, reliable communication enabled by 5G.

115. Next Generation Network (NGN) is primarily based on ………

a) Circuit-switching
b) Packet-switched IP networks
c) Morse code
d) Analog communication
Answer:
b) Packet-switched IP networks

Explanation:
NGN uses IP packet switching to carry voice, video, and data.

116. NGN is designed to integrate ………

a) Only audio communication
b) Multiple services like voice, video, and data
c) Only video streaming
d) Only SMS services
Answer:
b) Multiple services like voice, video, and data

Explanation:
NGN supports converged services over a unified IP network.

117. A key feature of NGN is ………

a) Service independence from the network
b) Limited bandwidth

c) Analog interface
d) No mobility support

Answer:
a) Service independence from the network

Explanation:
NGN separates service logic from the transport network for flexibility.

118. Which functional block of NGN controls call setup and teardown?

a) Media Gateway
b) Media Gateway Controller
c) Application Server
d) Router

Answer:
b) Media Gateway Controller

Explanation:
The Media Gateway Controller manages call control and signaling.

119. Media Gateway in NGN is responsible for ………

a) Converting media streams between networks
b) Performing user authentication
c) Controlling billing
d) Handling email traffic

Answer:
a) Converting media streams between networks

Explanation:
Media Gateways convert media formats between PSTN and IP networks.

120. The Application Server in NGN provides ………

a) Power supply
b) Service logic and applications
c) Radio frequency
d) Sensor data

Answer:
b) Service logic and applications

Explanation:
Application Servers host and manage service logic such as VoIP and multimedia services.

121. Which component translates between IP and traditional PSTN networks?

a) Router
b) Media Gateway
c) Application Server
d) GPS module

Answer:
b) Media Gateway

Explanation:
A Media Gateway converts media streams between IP-based networks and traditional PSTN networks.

122. NGN architecture supports mobility using ………

a) Static channels
b) Packet routing
c) Satellite radio
d) SMS center

Answer:
b) Packet routing

Explanation:
Packet routing allows seamless mobility by dynamically routing data across IP networks.

123. Which of these is a benefit of 5G in IoT applications?

a) Slow response time
b) Reduced device density
c) Massive connectivity
d) Weak security

Answer:
c) Massive connectivity

Explanation:
5G supports a very large number of connected IoT devices simultaneously.

124. The part of NGN that handles service creation is ………

a) Media Gateway
b) Application Server
c) PSTN switch
d) Antenna

Answer:
b) Application Server

Explanation:
The Application Server hosts service logic and manages application-level services.

125. 5G enables ultra-reliable communication also known as ………

a) URLLC
b) VoIP
c) FTP
d) SMS
Answer:
a) URLLC

Explanation:
URLLC stands for Ultra-Reliable Low-Latency Communication, a core feature of 5G.

126. NGN Media Gateway Controller communicates using ………

a) MGCP or H.248
b) PDF files
c) SMS protocol
d) Bluetooth
Answer:
a) MGCP or H.248

Explanation:
MGCP and H.248 are control protocols used by Media Gateway Controllers.

127. In 5G IoT, extremely low latency is essential for ………

a) Video buffering
b) Smart appliances
c) Industrial automation
d) File downloads

Answer:
c) Industrial automation

Explanation:
Industrial automation requires instant response and minimal delay.

128. The NGN architecture is often divided into ………

a) 1 layer
b) 2 layers
c) 3 layers
d) 7 layers

Answer:
c) 3 layers

Explanation:
NGN is commonly structured into transport, control, and service layers.

129. A key characteristic of NGN is ………

a) No support for multimedia
b) Quality of Service (QoS)
c) Static switching
d) Manual routing

Answer:
b) Quality of Service (QoS)

Explanation:
NGN supports QoS to ensure reliable delivery of voice, video, and data.

130. In NGN, the Media Gateway Controller controls the ………

a) Voice commands
b) Signaling messages
c) Laser signals
d) GSM only

Answer:
b) Signaling messages

Explanation:
The Media Gateway Controller manages signaling and call control.

131. Application servers in NGN are mainly ………

a) End-user devices
b) Service execution platforms
c) Physical switches
d) Radio towers

Answer:
b) Service execution platforms

Explanation:
They execute and manage service logic for NGN services.

132. 5G enhances IoT applications in healthcare by enabling ………

a) Offline medical tests
b) Remote patient monitoring
c) Manual data entry
d) Traditional telephones

Answer:
b) Remote patient monitoring

Explanation:
5G enables real-time health monitoring with low latency and high reliability.

133. Which feature of 5G helps in smart industrial robots?

a) High latency
b) Ultra-reliable low latency
c) Low bandwidth
d) Weak connectivity

Answer:
b) Ultra-reliable low latency

Explanation:
Industrial robots require precise and instant communication.

134. NGN supports which type of communication?

a) Circuit only
b) Multi-service packet communication
c) Manual signaling
d) Morse code

Answer:
b) Multi-service packet communication

Explanation:
NGN supports voice, video, and data over packet-switched IP networks.

135. The transport layer of NGN handles ....

a) Service creation
b) User traffic transmission
c) Application logic
d) Authentication only

Answer:
b) User traffic transmission

Explanation:
The transport layer carries media and data packets across the network.

136. The combination of 5G and IoT is mainly used for ....

a) Faster book printing
b) Automation and smart connectivity
c) Slower networks
d) Manual switching

Answer:
b) Automation and smart connectivity

Explanation:
5G + IoT enables intelligent automation across industries.

137. Cloud-based IoT architecture mainly integrates ....

a) Sensors and printers
b) IoT devices with cloud storage and computing
c) Only mobile phones
d) Offline storage systems

Answer:
b) IoT devices with cloud storage and computing

Explanation:
Cloud platforms provide storage, processing, and analytics for IoT data.

138. In cloud-based IoT, the role of the cloud is to ………

a) Only sense environmental data
b) Store, process, and analyze IoT data
c) Replace sensors
d) Provide audio output

Answer:
b) Store, process, and analyze IoT data

Explanation:
The cloud performs heavy computation and analytics for IoT systems.

139. Which layer of cloud-based IoT handles sensing and data collection?

a) Application layer
b) Sensor/Perception layer
c) Network management
d) Virtualization layer

Answer:
b) Sensor/Perception layer

Explanation:
This layer includes sensors and actuators that collect physical data.

140. In cloud-based IoT architecture, the network layer is responsible for ………

a) Data transmission between devices and cloud
b) Data visualization
c) Executing cloud programs
d) Powering devices

Answer:
a) Data transmission between devices and cloud

Explanation:
The network layer ensures reliable communication between IoT devices and cloud services.

141. Which cloud service model is commonly used for IoT data analytics?

a) IaaS
b) PaaS
c) SaaS

d) CDP

Answer:
b) PaaS

Explanation:
PaaS provides platforms and tools for IoT data processing and analytics.

142. The application layer in cloud-based IoT provides ………

a) Machine learning analysis
b) User-specific services like dashboards and alerts
c) Power supply management
d) Hardware manufacturing

Answer:
b) User-specific services like dashboards and alerts

Explanation:
The application layer interacts with users through interfaces and alerts.

143. Which of the following is NOT a component of cloud-based IoT architecture?

a) Perception layer
b) Network layer
c) Application layer
d) Manual switching layer

Answer:
d) Manual switching layer

Explanation:
Manual switching is not part of automated IoT architectures.

144. Cloud-based IoT improves scalability by ………

a) Restricting device connections
b) Allowing flexible storage and resources
c) Reducing storage options
d) Turning off cloud servers

Answer:
b) Allowing flexible storage and resources

Explanation:

Cloud resources can scale up or down based on demand.

145. Which cloud platform is widely used for IoT?

a) Adobe Photoshop
b) Microsoft Azure IoT Hub
c) VLC Media Player
d) GIMP

Answer:
b) Microsoft Azure IoT Hub

Explanation:
Azure IoT Hub is a popular cloud platform for IoT device management.

146. The main benefit of integrating IoT with cloud is ………

a) Manual data processing
b) Large-scale storage and real-time processing
c) Reduced connectivity
d) Less device automation

Answer:
b) Large-scale storage and real-time processing

Explanation:
Cloud computing enables efficient handling of massive IoT data.

147. Which layer in cloud-based IoT architecture is responsible for device connectivity and data acquisition?

a) Application layer
b) Perception layer
c) Network layer
d) Business layer

Answer:
b) Perception layer

Explanation:
The perception layer collects data from sensors and devices.

148. What is the main purpose of the Cloud Service Layer in IoT architecture?

a) Storing and processing IoT data
b) Connecting IoT devices to sensors
c) End-user interaction
d) Powering IoT hardware

Answer:
a) Storing and processing IoT data

Explanation:
Cloud service layer handles data storage, processing, and analytics.

149. In a cloud-IoT system, which component typically performs real-time analytics?

a) IoT device
b) Cloud platform
c) Local router
d) End user application

Answer:
b) Cloud platform

Explanation:
Cloud platforms provide high computing power for real-time analytics.

150. Which cloud deployment model is most suitable for organizations wanting full control over IoT data security?

a) Public Cloud
b) Private Cloud
c) Hybrid Cloud
d) Community Cloud

Answer:
b) Private Cloud

Explanation:
Private cloud offers maximum control and security over organizational data.

151. Who invented the term Internet of Things?
a) Bill Gates
b) Tim Berners-Lee
c) Kevin Ashton
d) Steve Jobs
Answer: c) Kevin Ashton

152.IoT devices are mainly embedded with:
a) Screens only
b) Sensors and software
c) Keyboards
d) Printers
Answer: b) Sensors and software

153.IoT mainly enables:
a) Offline communication
b) Manual systems
c) Interconnected smart devices
d) Only wired networking
Answer: c) Interconnected smart devices

154.RFID stands for:
a) Radio Frequency Internet Device
b) Radio Frequency Identification
c) Rapid File Identification
d) Real Frequency Data
Answer: b) Radio Frequency Identification

155.Which is NOT a characteristic of IoT?
a) Connectivity
b) Automation
c) Scalability
d) Isolation
Answer: d) Isolation

156.IoT allows automation to:
a) Increase manual work
b) Reduce human intervention
c) Stop communication
d) Increase delays
Answer: b) Reduce human intervention

157.Real-time processing is important in:
a) Smart IoT systems
b) Typewriters
c) Old telephone systems
d) Fax machines
Answer: a) Smart IoT systems

158.Indoor air quality monitoring is an example of:
a) Smart transport
b) IoT in Industry
c) Smart classroom

d) IoT gaming
Answer: b) IoT in Industry

159.Smart homes can be controlled using:
a) Calculator
b) Smart phone
c) Radio
d) Pager
Answer: b) Smart phone

160.IoT in agriculture helps in:
a) Gaming
b) Crop monitoring
c) Cinema
d) Banking only
Answer: b) Crop monitoring

161.IoT healthcare helps in:
a) Remote patient monitoring
b) Only surgery
c) File storage
d) Entertainment
Answer: a) Remote patient monitoring

162.GPS data in transport helps reduce:
a) Internet usage
b) Traffic congestion
c) Electricity
d) Agriculture production
Answer: b) Traffic congestion

163.IoT enables:
a) Manual communication
b) Device-to-device interaction
c) No connectivity
d) Paper-based work
Answer: b) Device-to-device interaction

164.Scalability means:
a) Fixed system
b) Growing system capability
c) Small storage
d) Low speed
Answer: b) Growing system capability

165.One major advantage of IoT is:
a) Slow processing
b) Cost efficiency

c) Manual control
d) Data loss
Answer: b) Cost efficiency

166.One disadvantage of IoT is:
a) Automation
b) Security vulnerability
c) Connectivity
d) Monitoring
Answer: b) Security vulnerability

167.IoT dependency mainly relies on:
a) Electricity only
b) Internet connectivity
c) Books
d) Paper
Answer: b) Internet connectivity

168.Data breaches in IoT affect:
a) Entertainment
b) Privacy and trust
c) Printing
d) Television
Answer: b) Privacy and trust

169.Physical design of IoT includes:
a) Sensors and protocols
b) Only cables
c) Only routers
d) Only processors
Answer: a) Sensors and protocols

170.Ethernet standard is:
a) 802.3
b) 802.11
c) 802.15
d) 802.16
Answer: a) 802.3

171.Wi-Fi standard belongs to:
a) 802.3
b) 802.11
c) 802.15.4
d) 802.16
Answer: b) 802.11

172.6LOWPAN works with:
a) 802.15.4

b) 802.3
c) 802.11
d) 4G
Answer: a) 802.15.4

173.IPv4 uses:
a) 32-bit address
b) 64-bit
c) 128-bit
d) 16-bit
Answer: a) 32-bit

174.IPv6 uses:
a) 32-bit
b) 64-bit
c) 128-bit
d) 256-bit
Answer: c) 128-bit

175.TCP is:
a) Connectionless
b) Connection oriented
c) Stateless
d) Wireless
Answer: b) Connection oriented

176.UDP is:
a) Reliable protocol
b) Connectionless protocol
c) Wired only
d) Slow protocol
Answer: b) Connectionless protocol

177.MQTT follows:
a) Request-response
b) Publish-subscribe
c) Push-pull
d) Exclusive pair
Answer: b) Publish-subscribe

178.HTTP works on:
a) UDP
b) TCP
c) Bluetooth
d) Zigbee
Answer: b) TCP

179.CoAP runs over:
a) TCP
b) UDP
c) FTP
d) SMTP
Answer: b) UDP

180.WebSocket supports:
a) Half duplex
b) Full duplex
c) Offline mode
d) File system
Answer: b) Full duplex

181.Sensor detects:
a) Physical parameters
b) Electricity bills
c) Software bugs
d) Internet speed
Answer: a) Physical parameters

182.Actuator performs:
a) Data storage
b) Physical action
c) Coding
d) Analysis
Answer: b) Physical action

183.Temperature sensor example:
a) DHT11
b) MQ-2
c) PIR
d) IR
Answer: a) DHT11

184.PIR is used for:
a) Temperature
b) Motion detection
c) Pressure
d) Light
Answer: b) Motion detection

185.BMP280 measures:
a) Gas
b) Pressure
c) Motion

d) Light
Answer: b) Pressure

186. MQ-2 detects:
a) Temperature
b) Gas
c) Humidity
d) Motion
Answer: b) Gas

187. Soil moisture sensor used in:
a) Industry
b) Agriculture
c) Banking
d) Cinema
Answer: b) Agriculture

188. Servo motor provides:
a) Linear motion
b) Precise angular position
c) Heating
d) Storage
Answer: b) Precise angular position

189. Relay acts as:
a) Sensor
b) Switch
c) Display
d) Processor
Answer: b) Switch

190. Solenoid provides:
a) Rotation
b) Linear motion
c) Light
d) Data
Answer: b) Linear motion

191. IoT gateway performs:
a) Data deletion
b) Pre-processing
c) Printing
d) Gaming
Answer: b) Pre-processing

192. Cloud layer provides:
a) Device sensing
b) Data storage and processing

c) Only wiring
d) Printing
Answer: b) Data storage and processing

193.SaaS stands for:
a) Software as a Service
b) Storage as System
c) Server as Software
d) System as Service
Answer: a) Software as a Service

194.IaaS provides:
a) Applications only
b) Infrastructure resources
c) Music
d) Books
Answer: b) Infrastructure resources

195.PaaS provides:
a) Platform for development
b) Hardware only
c) Only cables
d) Storage only
Answer: a) Platform for development

196.Exclusive pair model maintains:
a) Stateless connection
b) Persistent connection
c) No communication
d) Broadcast
Answer: b) Persistent connection

197.Publish-subscribe uses:
a) Broker
b) Printer
c) Cable
d) Router only
Answer: a) Broker

198.IoT security includes:
a) Authorization
b) Authentication
c) Encryption
d) All of the above
Answer: d) All of the above

199.LR-WPAN provides:
a) High speed

b) Low-rate wireless communication
c) Satellite link
d) Fiber
Answer: b) Low-rate wireless communication

200.Zigbee is based on:
a) 802.15.4
b) 802.3
c) 802.11
d) LTE
Answer: a) 802.15.4

201.IoT devices mainly communicate using:
a) Postal service
b) Internet protocols
c) Paper signals
d) Television waves
Answer: b) Internet protocols

202.Which layer handles host addressing and routing?
a) Application layer
b) Transport layer
c) Network layer
d) Physical layer
Answer: c) Network layer

203.IPv4 allows approximately how many addresses?
a) 1 million
b) 4.3 billion
c) 1 trillion
d) 100 million
Answer: b) 4.3 billion

204.6LOWPAN is mainly used for:
a) High-speed gaming
b) Low-power IoT devices
c) Video streaming
d) Satellite TV
Answer: b) Low-power IoT devices

205.Which protocol guarantees reliable data transfer?
a) UDP
b) MQTT
c) TCP
d) CoAP
Answer: c) TCP

206.UDP is suitable for:
a) File transfer
b) Email
c) Time-sensitive applications
d) Secure banking
Answer: c) Time-sensitive applications

207.HTTP follows which model?
a) Publish-subscribe
b) Push-pull
c) Request-response
d) Exclusive pair
Answer: c) Request-response

208.MQTT uses a:
a) Server only
b) Broker
c) Switch
d) Router only
Answer: b) Broker

209.CoAP is mainly designed for:
a) Large servers
b) Constrained devices
c) Supercomputers
d) Satellite communication
Answer: b) Constrained devices

210.WebSocket maintains:
a) Temporary link
b) Persistent connection
c) No connection
d) One-way communication
Answer: b) Persistent connection

211.IEEE 802.16 is related to:
a) WiFi
b) WiMax
c) Bluetooth
d) Zigbee
Answer: b) WiMax

212.IEEE 802.15.4 is used in:
a) Ethernet
b) Zigbee
c) LTE

d) Satellite
Answer: b) Zigbee

213.Transport layer provides:
a) Error control
b) Segmentation
c) Flow control
d) All of the above
Answer: d) All of the above

214.Port 80 is used by:
a) FTP
b) HTTP
c) SSH
d) SMTP
Answer: b) HTTP

215.MQTT is best suited for:
a) High bandwidth networks
b) Low bandwidth networks
c) Fiber optic only
d) Cable TV
Answer: b) Low bandwidth networks

216.XMPP is used for:
a) Real-time messaging
b) File compression
c) Hardware repair
d) Data storage
Answer: a) Real-time messaging

217.DDS stands for:
a) Data Distribution Service
b) Digital Data System
c) Device Data Storage
d) Direct Distribution Service
Answer: a) Data Distribution Service

218.AMQP is mainly used for:
a) Gaming
b) Business messaging
c) Image editing
d) Audio streaming
Answer: b) Business messaging

219.A sensor converts:
a) Digital signals to light
b) Physical signals to digital data

c) Electricity to gas
d) Data to paper
Answer: b) Physical signals to digital data

220.An actuator converts:
a) Data into physical action
b) Light into heat
c) Air into sound
d) Temperature into gas
Answer: a) Data into physical action

221.DHT22 measures:
a) Pressure
b) Humidity and temperature
c) Motion
d) Gas
Answer: b) Humidity and temperature

222.PIR sensors detect:
a) Sound
b) Motion
c) Light
d) Pressure
Answer: b) Motion

223.LDR is used for measuring:
a) Pressure
b) Gas
c) Light intensity
d) Motion
Answer: c) Light intensity

224.TPMS in vehicles monitors:
a) Fuel
b) Tyre pressure
c) Temperature
d) Engine speed
Answer: b) Tyre pressure

225.MQ series sensors detect:
a) Sound
b) Gas
c) Motion
d) Pressure
Answer: b) Gas

226.Accelerometer detects:
a) Light

b) Temperature
c) Orientation and vibration
d) Humidity
Answer: c) Orientation and vibration

227.IR sensors detect:
a) Heat radiation
b) Sound
c) Water
d) Pressure
Answer: a) Heat radiation

228.Servo motors are used for:
a) Heating
b) Precise angular movement
c) Sound
d) Light
Answer: b) Precise angular movement

229.Stepper motors move in:
a) Continuous rotation
b) Random motion
c) Discrete steps
d) Linear push
Answer: c) Discrete steps

230.DC motors convert:
a) Mechanical to electrical
b) Electrical to mechanical
c) Gas to liquid
d) Light to heat
Answer: b) Electrical to mechanical

231.A relay is mainly used to:
a) Sense motion
b) Switch high power circuits
c) Store data
d) Display output
Answer: b) Switch high power circuits

232.Solenoids are mainly used for:
a) Display
b) Linear push/pull action
c) Light sensing
d) Audio
Answer: b) Linear push/pull action

233.Cloud computing helps IoT in:
a) Data storage
b) Processing
c) Analytics
d) All of the above
Answer: d) All of the above

234.SaaS means:
a) Service as a Software
b) Software as a Service
c) Server as System
d) Storage as Software
Answer: b) Software as a Service

235.IaaS provides:
a) Applications
b) Platform
c) Infrastructure resources
d) Messaging
Answer: c) Infrastructure resources

236.PaaS is mainly used by:
a) Farmers
b) Developers
c) Drivers
d) Teachers
Answer: b) Developers

237.IoT gateway acts as:
a) Final storage
b) Intermediary device
c) Sensor
d) Actuator
Answer: b) Intermediary device

238.Data ingestion occurs in:
a) Device layer
b) Cloud platform layer
c) Sensor layer
d) Physical layer
Answer: b) Cloud platform layer

239.Application layer in IoT provides:
a) User interface
b) Wiring
c) Voltage

d) Hardware
Answer: a) User interface

240.Connectivity in IoT includes:
a) HTTP
b) MQTT
c) AMQP
d) All of the above
Answer: d) All of the above

241.Edge computing reduces:
a) Storage
b) Latency
c) Power
d) Sensors
Answer: b) Latency

242.IoT automation improves:
a) Manual errors
b) Efficiency
c) Delays
d) Cost increase
Answer: b) Efficiency

243.Security in IoT includes:
a) Encryption
b) Authentication
c) Authorization
d) All of the above
Answer: d) All of the above

244.Scalability ensures:
a) Fixed size system
b) System growth capability
c) Limited devices
d) No updates
Answer: b) System growth capability

245.Real-time IoT systems require:
a) Slow processing
b) Fast data processing
c) No internet
d) Manual control
Answer: b) Fast data processing

246.IoT in healthcare helps in:
a) Remote diagnosis
b) Entertainment

c) Printing
d) Banking
Answer: a) Remote diagnosis

247.Smart irrigation uses:
a) Soil moisture sensor
b) Sound sensor
c) Gas sensor
d) Light only
Answer: a) Soil moisture sensor

248.Interoperability means:
a) Devices work together
b) Devices isolated
c) No connectivity
d) Manual process
Answer: a) Devices work together

249.IoT devices generate:
a) Paper
b) Large amounts of data
c) Noise
d) Electricity only
Answer: b) Large amounts of data

250.Main goal of IoT is:
a) Isolation
b) Smart connectivity and automation
c) Manual processing
d) Offline storage
Answer: b) Smart connectivity and automation

251.IoT mainly connects:
a) People only
b) Devices only
c) Physical objects over Internet
d) Computers only
Answer: c) Physical objects over Internet

252.The main role of sensors in IoT is:
a) Actuation
b) Data collection
c) Data deletion
d) Data encryption
Answer: b) Data collection

253.Automation in IoT helps to:
a) Increase manual errors

b) Reduce human effort
c) Stop monitoring
d) Disable connectivity
Answer: b) Reduce human effort

254.IoT systems must support:
a) Interoperability
b) Isolation
c) Manual switching
d) Offline processing
Answer: a) Interoperability

255.CAN Bus is used for:
a) Wireless TV
b) Industrial communication
c) Satellite TV
d) FM radio
Answer: b) Industrial communication

256.Smart home devices communicate via:
a) Internet
b) Fax
c) Telegraph
d) Walkie-talkie
Answer: a) Internet

257.IoT agriculture systems analyze:
a) Weather and crop conditions
b) Cinema ratings
c) Stock exchange
d) Television signals
Answer: a) Weather and crop conditions

258.Remote monitoring is possible using:
a) IoT healthcare systems
b) Typewriters
c) Analog phones
d) Radio sets
Answer: a) IoT healthcare systems

259.IoT transportation systems use:
a) GPS
b) FM
c) CD player
d) Pager
Answer: a) GPS

260.One key feature of IoT is:
a) Real-time capability
b) Manual updates
c) Slow speed
d) Fixed size
Answer: a) Real-time capability

261.Ethernet operates at which layer?
a) Application
b) Transport
c) Link layer
d) Session
Answer: c) Link layer

262.802.11 operates in:
a) Wireless LAN
b) Satellite
c) Cable TV
d) Telephone
Answer: a) Wireless LAN

263.802.15.4 supports:
a) High bandwidth
b) Low power devices
c) Satellite link
d) 5G
Answer: b) Low power devices

264.4G data rate can reach up to:
a) 10 Mbps
b) 100 Mbps
c) 100 Mbps+
d) 1 Kbps
Answer: c) 100 Mbps+

265.Network layer performs:
a) Packet routing
b) File editing
c) Image processing
d) Hardware repair
Answer: a) Packet routing

266.IPv6 supports:
a) 32-bit addressing
b) 128-bit addressing
c) 16-bit addressing

d) 8-bit addressing
Answer: b) 128-bit addressing

267.Transport layer ensures:
a) End-to-end communication
b) Light intensity
c) Power supply
d) Screen display
Answer: a) End-to-end communication

268.TCP provides:
a) Unreliable transfer
b) Reliable transfer
c) No error control
d) No sequencing
Answer: b) Reliable transfer

269.UDP does NOT provide:
a) Guaranteed delivery
b) Speed
c) Low overhead
d) Fast transmission
Answer: a) Guaranteed delivery

270.HTTP uses:
a) Port 80
b) Port 22
c) Port 21
d) Port 25
Answer: a) Port 80

271.SSH uses port:
a) 80
b) 22
c) 21
d) 110
Answer: b) 22

272.MQTT stands for:
a) Message Queue Telemetry Transport
b) Multi Queue Transfer Tool
c) Modern Query Transmission Tool
d) Message Quick Transport Technology
Answer: a) Message Queue Telemetry Transport

273.CoAP is mainly used for:
a) Constrained environments
b) Video streaming

c) High gaming
d) 5G core
Answer: a) Constrained environments

274.Publish-subscribe model includes:
a) Publisher
b) Broker
c) Subscriber
d) All of the above
Answer: d) All of the above

275.Push-pull model uses:
a) Queue system
b) Email system
c) Paper system
d) Fax
Answer: a) Queue system

276.Exclusive pair model is:
a) Half duplex
b) Full duplex
c) Offline
d) Broadcast
Answer: b) Full duplex

277.Data analytics in IoT helps in:
a) Deriving insights
b) Deleting files
c) Printing data
d) Shutting down devices
Answer: a) Deriving insights

278.Scalability means system should:
a) Remain fixed
b) Grow with devices
c) Decrease devices
d) Stop updates
Answer: b) Grow with devices

279.IoT systems require:
a) Energy efficiency
b) High power consumption
c) No connectivity
d) No sensors
Answer: a) Energy efficiency

280.One disadvantage of IoT is:
a) Improved monitoring

b) Overreliance on technology
c) Automation
d) Connectivity
Answer: b) Overreliance on technology

281.Security vulnerabilities in IoT may cause:
a) Data theft
b) Faster speed
c) Better monitoring
d) No risk
Answer: a) Data theft

282.Operational complexity refers to:
a) Easy backend
b) Complex backend processes
c) No coding
d) Manual system
Answer: b) Complex backend processes

283.IoT reduces employment mainly in:
a) High skilled jobs
b) Low skilled repetitive jobs
c) Doctors
d) Engineers
Answer: b) Low skilled repetitive jobs

284.IoT gateway connects:
a) Devices and cloud
b) Printer and cable
c) Paper and file
d) CD and DVD
Answer: a) Devices and cloud

285.Device layer contains:
a) Sensors and actuators
b) Servers only
c) Printers
d) Routers
Answer: a) Sensors and actuators

286.Network layer ensures:
a) Data transmission
b) Cooking
c) Printing
d) Gaming
Answer: a) Data transmission

287.Cloud platform provides:
a) Storage
b) Processing
c) Device management
d) All of the above
Answer: d) All of the above

288.Application layer provides:
a) User interface
b) Wiring
c) Voltage
d) Temperature
Answer: a) User interface

289.Soil moisture sensors are used in:
a) Smart farming
b) Gaming
c) Banking
d) Cinema
Answer: a) Smart farming

290.Gas sensors are used for:
a) Air quality monitoring
b) Temperature only
c) Motion only
d) Light only
Answer: a) Air quality monitoring

291.Humidity sensors measure:
a) Water vapour in air
b) Heat
c) Light
d) Sound
Answer: a) Water vapour in air

292.Light sensors measure:
a) Illumination
b) Gas
c) Pressure
d) Motion
Answer: a) Illumination

293.Accelerometer is used in:
a) Smartphones
b) Calculators
c) Radios

d) Fax
Answer: a) Smartphones

294.Actuators in irrigation control:
a) Water pumps
b) Television
c) Radio
d) Printer
Answer: a) Water pumps

295.Solenoid works using:
a) Electromagnetism
b) Gravity
c) Wind
d) Solar panel
Answer: a) Electromagnetism

296.Relay can control:
a) High power AC circuits
b) Temperature
c) Gas
d) Light intensity
Answer: a) High power AC circuits

297.Edge computing helps reduce:
a) Latency
b) Speed
c) Storage
d) Accuracy
Answer: a) Latency

298.IoT analytics supports:
a) Decision making
b) File deletion
c) Manual errors
d) Power cut
Answer: a) Decision making

299.Smart healthcare devices monitor:
a) Chronic conditions
b) Weather
c) Traffic
d) Agriculture
Answer: a) Chronic conditions

300.Main objective of IoT design is:
a) Connectivity + Intelligence
b) Isolation

c) Manual work
d) Offline operation
Answer: a) Connectivity + Intelligence

301.The physical design of IoT includes:
a) Devices and protocols
b) Only software
c) Only hardware
d) Only cloud
Answer: a) Devices and protocols

302.Logical design of IoT includes:
a) Functional blocks
b) Communication models
c) APIs
d) All of the above
Answer: d) All of the above

303.IoT functional blocks include:
a) Devices
b) Communication
c) Security
d) All of the above
Answer: d) All of the above

304.Device functional block performs:
a) Sensing and actuation
b) Routing
c) Printing
d) Gaming
Answer: a) Sensing and actuation

305.Security block provides:
a) Authentication
b) Authorization
c) Data security
d) All of the above
Answer: d) All of the above

306.Management block is responsible for:
a) System control and configuration
b) Motion detection
c) Light sensing
d) Pressure control
Answer: a) System control and configuration

307.Service block provides:
a) Monitoring and control

b) Video editing
c) Gaming
d) Television
Answer: a) Monitoring and control

308.Communication block handles:
a) Data exchange
b) Heating
c) Cooling
d) Storage only
Answer: a) Data exchange

309.Application block allows users to:
a) Monitor system
b) Analyze data
c) Control devices
d) All of the above
Answer: d) All of the above

310.Request-response model is:
a) Stateless
b) Stateful
c) Offline
d) Broadcast
Answer: a) Stateless

311.In request-response model, client sends:
a) Signal
b) Request
c) Packet
d) File
Answer: b) Request

312.Publish-subscribe model separates:
a) Publisher and subscriber
b) Sensor and actuator
c) User and device
d) Cloud and network
Answer: a) Publisher and subscriber

313.In publish-subscribe model, broker manages:
a) Topics
b) Hardware
c) Voltage
d) Electricity
Answer: a) Topics

314.Push-pull model uses:
a) Queue
b) Router
c) Switch
d) Printer
Answer: a) Queue

315.Exclusive pair model is also called:
a) Persistent connection model
b) Stateless model
c) Offline model
d) Analog model
Answer: a) Persistent connection model

316.REST APIs use:
a) HTTP methods
b) Bluetooth
c) FM signals
d) Coaxial cable
Answer: a) HTTP methods

317.REST commonly uses:
a) GET
b) POST
c) PUT
d) All of the above
Answer: d) All of the above

318.WebSocket is suitable for:
a) Real-time dashboards
b) Printing
c) Storage
d) Offline apps
Answer: a) Real-time dashboards

319.MQTT is designed for:
a) High-power devices
b) Low-power devices
c) Gaming consoles
d) Television
Answer: b) Low-power devices

320.CoAP is similar to:
a) FTP
b) HTTP
c) SMTP

d) SSH
Answer: b) HTTP

321.AMQP supports:
a) Point-to-point messaging
b) Publish-subscribe
c) Routing
d) All of the above
Answer: d) All of the above

322.XMPP is based on:
a) XML
b) JSON only
c) Binary code
d) Image format
Answer: a) XML

323.Link layer protocols determine:
a) Physical transmission method
b) Application logic
c) User interface
d) Cloud storage
Answer: a) Physical transmission method

324.802.3 refers to:
a) Ethernet
b) WiFi
c) Zigbee
d) LTE
Answer: a) Ethernet

325.802.11 refers to:
a) WiFi
b) Ethernet
c) Bluetooth
d) 5G
Answer: a) WiFi

326.802.15.4 provides data rate up to:
a) 1 Mbps
b) 250 Kbps
c) 10 Mbps
d) 1 Gbps
Answer: b) 250 Kbps

327.2G data rate starts from:
a) 9.6 Kbps
b) 100 Mbps

c) 1 Gbps
d) 10 Mbps
Answer: a) 9.6 Kbps

328.Network layer addressing uses:
a) IP address
b) MAC only
c) Port only
d) Cable ID
Answer: a) IP address

329.TCP ensures:
a) Packet ordering
b) Error detection
c) Retransmission
d) All of the above
Answer: d) All of the above

330.UDP is suitable for:
a) Streaming
b) Gaming
c) VoIP
d) All of the above
Answer: d) All of the above

331.HTTP is:
a) Stateless protocol
b) Stateful protocol
c) Connectionless only
d) Hardware protocol
Answer: a) Stateless protocol

332.CoAP runs on:
a) TCP
b) UDP
c) Bluetooth
d) Cable
Answer: b) UDP

333.MQTT architecture is:
a) Client-broker
b) Client-only
c) Peer-to-peer only
d) Router-based
Answer: a) Client-broker

334.DDS provides:
a) QoS control

b) Gaming
c) Printing
d) Analog signaling
Answer: a) QoS control

335.Data ingestion refers to:
a) Receiving device data
b) Deleting files
c) Formatting disk
d) Repairing hardware
Answer: a) Receiving device data

336.Device management in cloud handles:
a) Provisioning
b) Identity
c) Remote management
d) All of the above
Answer: d) All of the above

337.Edge functions manage:
a) Traffic aggregation
b) File printing
c) Voltage
d) Audio
Answer: a) Traffic aggregation

338.Core transport functions ensure:
a) High capacity routing
b) Manual switching
c) Heating
d) Lighting
Answer: a) High capacity routing

339.Gateway functions provide:
a) Interoperability
b) Motion
c) Gas detection
d) Light
Answer: a) Interoperability

340.NACF handles:
a) User authentication
b) File storage
c) Sensor design
d) Video editing
Answer: a) User authentication

341.RACF manages:
a) Resource allocation
b) Light intensity
c) Temperature
d) Sound
Answer: a) Resource allocation

342.IoT scalability allows:
a) Addition of devices
b) Removal of internet
c) Stop updates
d) Disable sensors
Answer: a) Addition of devices

343.Smart irrigation example shows:
a) Sensing + actuation
b) Only sensing
c) Only storage
d) Only cloud
Answer: a) Sensing + actuation

344.IoT analytics converts data into:
a) Insights
b) Paper
c) Noise
d) Errors
Answer: a) Insights

345.Real-time IoT systems require:
a) Low latency
b) High delay
c) No internet
d) Manual control
Answer: a) Low latency

346.Interoperability problem occurs due to:
a) Different manufacturers
b) Too much power
c) Excess storage
d) Too much speed
Answer: a) Different manufacturers

347.Energy-efficient IoT devices help in:
a) Longer battery life
b) High power use
c) Short life

d) More heat
Answer: a) Longer battery life

348.Cloud-based IoT architecture has:
a) Device layer
b) Network layer
c) Cloud layer
d) All of the above
Answer: d) All of the above

349.IoT mainly connects digital world with:
a) Physical world
b) Cinema
c) Music
d) Books
Answer: a) Physical world

350.The ultimate aim of IoT systems is:
a) Intelligent automation
b) Manual processing
c) Isolation
d) No connectivity
Answer: a) Intelligent automation

351.IoT enables communication between:
a) Human to human only
b) Machine to machine
c) Paper to printer
d) TV to radio
Answer: b) Machine to machine

352.Smart city applications include:
a) Traffic monitoring
b) Waste management
c) Smart lighting
d) All of the above
Answer: d) All of the above

353.In IoT architecture, the first layer is:
a) Cloud layer
b) Device layer
c) Application layer
d) Network layer
Answer: b) Device layer

354.IoT data analytics helps in:
a) Predictive analysis
b) Manual logging

c) Deleting files
d) Formatting
Answer: a) Predictive analysis

355.Low latency is important in:
a) Real-time systems
b) Offline systems
c) Manual systems
d) Paper systems
Answer: a) Real-time systems

356.IoT devices require:
a) Unique identification
b) Common name
c) No address
d) Paper code
Answer: a) Unique identification

357.A MAC address is used at:
a) Application layer
b) Network layer
c) Link layer
d) Cloud layer
Answer: c) Link layer

358.IP address is used at:
a) Physical layer
b) Network layer
c) Session layer
d) Presentation layer
Answer: b) Network layer

359.IoT gateways perform protocol:
a) Translation
b) Deletion
c) Printing
d) Heating
Answer: a) Translation

360.Fog computing is used to:
a) Increase latency
b) Reduce latency
c) Remove internet
d) Disable devices
Answer: b) Reduce latency

361.IoT applications generate:
a) Structured data only

b) Unstructured data only
c) Large variety of data
d) No data
Answer: c) Large variety of data

362.Cloud analytics provides:
a) Data visualization
b) Data processing
c) Data storage
d) All of the above
Answer: d) All of the above

363.IoT security must protect:
a) Data confidentiality
b) Data integrity
c) Data availability
d) All of the above
Answer: d) All of the above

364.Encryption ensures:
a) Data secrecy
b) Faster speed
c) More storage
d) Low power
Answer: a) Data secrecy

365.Authentication verifies:
a) Device identity
b) Voltage
c) Temperature
d) Pressure
Answer: a) Device identity

366.Authorization determines:
a) Who can access resources
b) Power supply
c) Sensor type
d) Cable type
Answer: a) Who can access resources

367.IoT scalability means:
a) Fixed device count
b) Expandable network
c) No updates
d) Low power
Answer: b) Expandable network

368.Sensor calibration improves:
a) Accuracy
b) Speed
c) Storage
d) Memory
Answer: a) Accuracy

369.Edge devices process data:
a) Near data source
b) Only in cloud
c) In server room
d) Offline
Answer: a) Near data source

370.MQTT topics are used for:
a) Data categorization
b) Cable management
c) Voltage control
d) Screen display
Answer: a) Data categorization

371.IoT in smart grid helps in:
a) Power monitoring
b) Manual billing
c) Paper records
d) Offline payment
Answer: a) Power monitoring

372.IoT wearable devices monitor:
a) Health parameters
b) Movies
c) Music
d) Games
Answer: a) Health parameters

373.Smart parking systems use:
a) Sensors
b) Routers
c) TV
d) Printer
Answer: a) Sensors

374.IoT devices must support:
a) Interoperability
b) Isolation
c) Offline work

d) Paper process
Answer: a) Interoperability

375.Big Data is related to IoT because:
a) IoT generates huge data
b) IoT deletes data
c) IoT blocks data
d) IoT stores paper
Answer: a) IoT generates huge data

376.Data visualization tools help in:
a) Graphical representation
b) Deletion
c) Formatting
d) Storage
Answer: a) Graphical representation

377.IoT device firmware is:
a) Embedded software
b) Printer driver
c) Game file
d) Music file
Answer: a) Embedded software

378.Remote firmware update improves:
a) Device functionality
b) Data loss
c) Speed reduction
d) Power loss
Answer: a) Device functionality

379.IoT testing ensures:
a) Reliability
b) Performance
c) Security
d) All of the above
Answer: d) All of the above

380.Smart meters are used for:
a) Energy measurement
b) Light only
c) Motion only
d) Gas only
Answer: a) Energy measurement

381.Zigbee is suitable for:
a) Short-range communication
b) Long-range satellite

c) Fiber optic
d) Television
Answer: a) Short-range communication

382.Bluetooth Low Energy is used for:
a) Wearables
b) Satellite
c) Fiber
d) Landline
Answer: a) Wearables

383.LPWAN stands for:
a) Low Power Wide Area Network
b) Large Power Wireless Access Network
c) Low Protocol WAN
d) Long Packet WAN
Answer: a) Low Power Wide Area Network

384.LoRaWAN is used for:
a) Long range IoT communication
b) Television
c) FM radio
d) Printer
Answer: a) Long range IoT communication

385.IoT architecture must ensure:
a) Reliability
b) Scalability
c) Security
d) All of the above
Answer: d) All of the above

386.Smart healthcare IoT reduces:
a) Hospital visits
b) Connectivity
c) Storage
d) Accuracy
Answer: a) Hospital visits

387.IoT automation improves:
a) Efficiency
b) Errors
c) Delay
d) Manual work
Answer: a) Efficiency

388.IoT device provisioning means:
a) Configuring devices

b) Deleting files
c) Printing
d) Gaming
Answer: a) Configuring devices

389.Data aggregation combines:
a) Multiple data sources
b) Files
c) Screens
d) Cables
Answer: a) Multiple data sources

390.IoT systems are part of:
a) Cyber-Physical Systems
b) Only software
c) Only hardware
d) Only networking
Answer: a) Cyber-Physical Systems

391.Quality of Service (QoS) ensures:
a) Performance guarantee
b) Slow speed
c) Data loss
d) Cable damage
Answer: a) Performance guarantee

392.Device authentication prevents:
a) Unauthorized access
b) Speed increase
c) More storage
d) Extra voltage
Answer: a) Unauthorized access

393.Cloud dashboard displays:
a) Real-time data
b) Hardware
c) Cable
d) Electricity
Answer: a) Real-time data

394.IoT data lifecycle includes:
a) Collection
b) Storage
c) Processing
d) All of the above
Answer: d) All of the above

395.IoT device battery life depends on:
a) Power efficiency
b) Cable length
c) Printer speed
d) Screen size
Answer: a) Power efficiency

396.IoT networks must handle:
a) Device heterogeneity
b) Single device
c) No device
d) Only PC
Answer: a) Device heterogeneity

397.Interoperability standards ensure:
a) Cross-platform compatibility
b) Isolation
c) Manual control
d) Paper communication
Answer: a) Cross-platform compatibility

398.Predictive maintenance uses:
a) Sensor data analysis
b) Printing
c) Formatting
d) Offline systems
Answer: a) Sensor data analysis

399.IoT data storage commonly uses:
a) Cloud databases
b) Paper files
c) Fax
d) TV
Answer: a) Cloud databases

400.The core benefit of IoT is:
a) Smart decision making
b) Manual work
c) Isolation
d) Offline processing
Answer: a) Smart decision making

401.IoT integrates the physical world with:
a) Digital systems
b) Paper systems
c) Analog radio

d) Manual tools
Answer: a) Digital systems

402.Smart lighting systems use:
a) Motion sensors
b) Typewriters
c) CDs
d) Printers
Answer: a) Motion sensors

403.IoT systems rely heavily on:
a) Data collection
b) Manual recording
c) Paper filing
d) Fax
Answer: a) Data collection

404.Low power consumption is important for:
a) Battery-operated IoT devices
b) Desktop PCs
c) Servers only
d) Printers
Answer: a) Battery-operated IoT devices

405.IoT systems generate value by:
a) Data analysis
b) Data deletion
c) Formatting
d) Printing
Answer: a) Data analysis

406.Device heterogeneity means:
a) Same type devices
b) Different types of devices
c) No devices
d) Only sensors
Answer: b) Different types of devices

407.Data transmission delay is called:
a) Bandwidth
b) Latency
c) Throughput
d) Voltage
Answer: b) Latency

408.IoT network reliability ensures:
a) Continuous service
b) Frequent breakdown

c) Manual repair
d) Offline mode
Answer: a) Continuous service

409.Data compression in IoT helps:
a) Reduce bandwidth usage
b) Increase errors
c) Slow speed
d) Increase latency
Answer: a) Reduce bandwidth usage

410.IoT firmware updates improve:
a) Security
b) Functionality
c) Performance
d) All of the above
Answer: d) All of the above

411.Edge analytics processes data:
a) At cloud only
b) Near source
c) In printer
d) In cable
Answer: b) Near source

412.IoT dashboards are used for:
a) Monitoring
b) Data visualization
c) Control
d) All of the above
Answer: d) All of the above

413.IoT automation reduces:
a) Operational cost
b) Electricity
c) Sensors
d) Network
Answer: a) Operational cost

414.Smart meters send data to:
a) Utility provider
b) Radio station
c) Printer
d) TV
Answer: a) Utility provider

415.IoT enhances productivity by:
a) Real-time insights

b) Manual logging
c) Offline systems
d) Paper tracking
Answer: a) Real-time insights

416.Data privacy laws affect:
a) IoT deployment
b) Cable type
c) Printer ink
d) TV signals
Answer: a) IoT deployment

417.An IoT actuator responds to:
a) Control signal
b) Voltage spike only
c) Printer command
d) Cable signal
Answer: a) Control signal

418.IoT scalability requires:
a) Flexible architecture
b) Fixed system
c) No expansion
d) Manual scaling
Answer: a) Flexible architecture

419.Secure IoT communication uses:
a) Encryption protocols
b) Paper logs
c) Offline systems
d) Manual approval
Answer: a) Encryption protocols

420.IoT smart locks use:
a) Sensors and internet
b) Printer
c) Typewriter
d) Radio
Answer: a) Sensors and internet

421.Industrial IoT improves:
a) Process efficiency
b) Manual work
c) Paper tracking
d) Delay
Answer: a) Process efficiency

422.IoT predictive analytics helps:
a) Anticipate failures
b) Increase downtime
c) Delete data
d) Reduce connectivity
Answer: a) Anticipate failures

423.IoT requires robust:
a) Network infrastructure
b) Paper
c) Manual system
d) Radio
Answer: a) Network infrastructure

424.IoT interoperability ensures:
a) Devices communicate seamlessly
b) Devices isolated
c) No integration
d) Manual data entry
Answer: a) Devices communicate seamlessly

425.Real-time IoT systems must handle:
a) Continuous data streams
b) Paper files
c) Fax
d) Typewriter
Answer: a) Continuous data streams

426.Cloud-based IoT allows:
a) Remote access
b) Offline only
c) Manual storage
d) No monitoring
Answer: a) Remote access

427.IoT remote monitoring improves:
a) Service quality
b) Errors
c) Downtime
d) Isolation
Answer: a) Service quality

428.IoT smart farming reduces:
a) Water waste
b) Productivity
c) Monitoring

d) Efficiency
Answer: a) Water waste

429.Data filtering helps:
a) Remove unnecessary data
b) Increase noise
c) Increase storage
d) Increase errors
Answer: a) Remove unnecessary data

430.IoT deployment challenges include:
a) Security issues
b) High cost
c) Integration complexity
d) All of the above
Answer: d) All of the above

431.IoT enhances smart cities by:
a) Intelligent traffic control
b) Manual signals
c) Paper tickets
d) Radio
Answer: a) Intelligent traffic control

432.IoT device lifecycle includes:
a) Deployment
b) Maintenance
c) Decommissioning
d) All of the above
Answer: d) All of the above

433.Smart irrigation systems use:
a) Soil sensors
b) TV
c) Printer
d) Fax
Answer: a) Soil sensors

434.IoT wearables communicate using:
a) Bluetooth
b) Fax
c) Telegraph
d) Analog radio
Answer: a) Bluetooth

435.IoT cloud platforms provide:
a) Analytics tools
b) Storage

c) Device management
d) All of the above
Answer: d) All of the above

436.Data encryption prevents:
a) Unauthorized access
b) Fast speed
c) Manual work
d) Storage increase
Answer: a) Unauthorized access

437.IoT edge computing reduces:
a) Bandwidth usage
b) Efficiency
c) Security
d) Connectivity
Answer: a) Bandwidth usage

438.IoT security must defend against:
a) Cyber attacks
b) Light
c) Heat
d) Gas
Answer: a) Cyber attacks

439.IoT big data requires:
a) Analytics tools
b) Paper storage
c) Typewriter
d) Fax
Answer: a) Analytics tools

440.Smart energy systems monitor:
a) Electricity consumption
b) Water only
c) Gas only
d) Paper usage
Answer: a) Electricity consumption

441.IoT communication protocols ensure:
a) Standardized data exchange
b) Isolation
c) Manual work
d) Offline processing
Answer: a) Standardized data exchange

442.IoT-enabled smart grids improve:
a) Energy distribution

b) Manual control
c) Paper billing
d) Delay
Answer: a) Energy distribution

443.IoT performance metrics include:
a) Latency
b) Throughput
c) Reliability
d) All of the above
Answer: d) All of the above

444.Device provisioning includes:
a) Initial configuration
b) Manual wiring
c) Deleting files
d) Printing
Answer: a) Initial configuration

445.IoT cloud computing supports:
a) Scalability
b) Flexibility
c) Resource optimization
d) All of the above
Answer: d) All of the above

446.IoT automation improves:
a) Accuracy
b) Consistency
c) Efficiency
d) All of the above
Answer: d) All of the above

447.IoT data storage commonly uses:
a) Databases
b) Paper
c) Radio
d) Printer
Answer: a) Databases

448.IoT integration requires:
a) Standard protocols
b) Isolation
c) No network
d) Manual entry
Answer: a) Standard protocols

449.IoT real-time alerts help in:

a) Immediate action
b) Delay
c) Isolation
d) Offline storage
Answer: a) Immediate action

450.The ultimate goal of IoT systems is:

a) Intelligent automation and decision making
b) Manual processes
c) Paper systems
d) Offline mode
Answer: a) Intelligent automation and decision making

b)

a)

UNIT 3 – BLOCKCHAIN TECHNOLOGY

1. Who invented blockchain technology?

a) Elon Musk
b) Satoshi Nakamoto
c) Vitalik Buterin
d) Tim Berners-Lee

Answer: b) Satoshi Nakamoto
Explanation: Blockchain technology was introduced by Satoshi Nakamoto in 2008 along with Bitcoin as a decentralized digital currency system.

2. What is a blockchain?

a) A centralized database
b) A decentralized digital ledger
c) A traditional banking system
d) A programming language

Answer: b) A decentralized digital ledger
Explanation: Blockchain is a distributed ledger that records transactions across multiple nodes without central authority.

3. What is the first block in the blockchain called?

a) Prime block
b) Genesis block
c) Root block
d) Initial block
Answer: b) Genesis block
Explanation: The first block of any blockchain is called the Genesis block and it has no previous block hash.

4. The blockchain ledger is _________.

a) Mutable
b) Immutable
c) Partially mutable
d) Deleted regularly
Answer: b) Immutable
Explanation: Once data is recorded in a blockchain, it cannot be changed or deleted, ensuring data integrity.

5. Which feature allows all nodes to keep a copy of the blockchain?

a) Transparency
b) Decentralization
c) Anonymity
d) Centralization
Answer: b) Decentralization
Explanation: In decentralization, every node maintains a copy of the blockchain ledger.

6. The hash in a blockchain block is ________.

a) The user's password
b) A unique identifier of the block
c) The blockchain's name
d) The transaction amount
Answer: b) A unique identifier of the block
Explanation: Hash uniquely identifies a block and ensures data integrity.

7. What links a block to its previous block?

a) Timestamp
b) Previous block's hash
c) User ID
d) Transaction amount
Answer: b) Previous block's hash
Explanation: Each block stores the hash of the previous block, forming a secure chain.

8. What process do miners perform?

a) Code smart contracts
b) Verify and add blocks to the blockchain
c) Delete invalid transactions
d) Create cryptocurrencies
Answer: b) Verify and add blocks to the blockchain
Explanation: Miners validate transactions and add them to the blockchain through consensus mechanisms.

9. Which of the following is NOT a key feature of blockchain?

a) Centralized control
b) Immutability
c) Transparency
d) Decentralization

Answer: a) Centralized control
Explanation: Blockchain is decentralized, not centrally controlled.

10. What ensures transactions on the blockchain cannot be altered?

a) Consensus algorithm
b) Digital wallets
c) Miners' IDs
d) Private keys

Answer: a) Consensus algorithm
Explanation: Consensus algorithms ensure agreement across nodes and prevent unauthorized modifications.

11. Blockchain network participants are called ________.

a) Servers
b) Miners
c) Nodes
d) Clients

Answer: c) Nodes
Explanation: Nodes are computers that participate in maintaining the blockchain network.

12. What property ensures all network copies remain consistent?

a) Central authority control
b) Consensus protocols
c) Data encryption
d) Transaction fees
Answer: b) Consensus protocols
Explanation: Consensus ensures all nodes agree on the same blockchain state.

13. Which system has a single point of failure?

a) Blockchain
b) Traditional centralized system
c) Peer-to-peer network
d) Decentralized system
Answer: b) Traditional centralized system
Explanation: Centralized systems rely on one authority, making them vulnerable to failure.

14. Which system provides higher transparency?

a) Traditional system
b) Blockchain
c) Neither
d) Both equally

Answer: b) Blockchain
Explanation: Blockchain transactions are visible and verifiable by all participants.

15. Which system is typically slower for cross-border transactions?

a) Blockchain
b) Traditional centralized system
c) Both the same
d) None of these

Answer: b) Traditional centralized system

Explanation: Traditional banking systems involve intermediaries and delays.

16. Blockchain eliminates the need for ________.

a) Cryptography
b) Intermediaries
c) Data storage
d) Recording transactions

Answer: b) Intermediaries
Explanation: Blockchain enables peer-to-peer transactions without third parties.

17. Public blockchains are ________.

a) Permissioned and private
b) Open to anyone
c) Operated by a single company
d) Always faster than private blockchains

Answer: b) Open to anyone
Explanation: Public blockchains allow anyone to join, read, and write transactions.

18. Private blockchains are ________.

a) Open to the public
b) Permissioned and controlled by an organization
c) Same as public blockchains

d) More decentralized

Answer: b) Permissioned and controlled by an organization

Explanation: Private blockchains restrict access and are managed by a single entity.

19. Consortium blockchains are ________.

a) Public blockchains
b) Governed by multiple organizations
c) Controlled by no one
d) Managed by governments

Answer: b) Governed by multiple organizations
Explanation: Consortium blockchains are jointly controlled by multiple institutions.

20. Hybrid blockchains ________.

a) Combine features of public and private blockchains
b) Are fully centralized
c) Are prohibited
d) Do not use cryptography

Answer: a) Combine features of public and private blockchains
Explanation: Hybrid blockchains provide controlled access with transparency benefits.

21. Which industry uses blockchain for secure cross-border payments?

a) Healthcare
b) Finance
c) Supply Chain
d) Gaming

Answer: b) Finance
Explanation: Blockchain enables fast and secure international financial transactions.

22. Blockchain improves supply chains by ________.

a) Reducing transparency
b) Tracking products from origin to consumer
c) Hiding product details
d) Increasing fraud risk

Answer: b) Tracking products from origin to consumer
Explanation: Blockchain ensures traceability and transparency in supply chains.

23. In healthcare, blockchain helps by ________.

a) Making patient data public
b) Securely storing and sharing medical records

c) Slowing data access
d) Reducing data collection

Answer: b) Securely storing and sharing medical records
Explanation: Blockchain improves security and interoperability of healthcare data.

24. Blockchain offers gamers the ability to ________.

a) Own in-game assets as NFTs
b) Cheat on games
c) Avoid transparency
d) Use centralized servers
Answer: a) Own in-game assets as NFTs
Explanation: NFTs allow true ownership of digital assets in gaming.

25. Smart contracts ________.

a) Require a lawyer
b) Are self-executing contracts coded on blockchain
c) Can be changed anytime
d) Need intermediaries
Answer: b) Are self-executing contracts coded on blockchain
Explanation: Smart contracts automatically execute when conditions are met.

26. Which is a key feature of smart contracts?

a) Autonomy and immutability
b) Manual execution
c) Human intervention
d) Easy erasure
Answer: a) Autonomy and immutability
Explanation: Smart contracts run independently and cannot be altered once deployed.

27. Bitcoin is ________.

a) A centralized digital currency
b) A cryptocurrency operating on blockchain
c) A fiat currency
d) Not related to blockchain
Answer: b) A cryptocurrency operating on blockchain
Explanation: Bitcoin was the first blockchain-based cryptocurrency.

28. Ethereum is known for ________.

a) Only cryptocurrency transactions
b) Enabling smart contract execution
c) Being private blockchain
d) Centralized control

Answer: b) Enabling smart contract execution
Explanation: Ethereum supports decentralized applications via smart contracts.

29. The biggest scalability challenge is ________.

a) Slow transactions as network grows
b) Too much transparency
c) Lack of security
d) Too many users
Answer: a) Slow transactions as network grows
Explanation: Increased network size affects processing speed.

30. Proof of Work consensus consumes ________.

a) Minimal energy
b) High energy due to mining computations
c) No electricity
d) Renewable resources only
Answer: b) High energy due to mining computations
Explanation: PoW requires heavy computational power, consuming significant energy.

31. Regulatory uncertainty ________.

a) Encourages rapid adoption
b) Hinders blockchain adoption globally
c) Is resolved everywhere
d) Does not affect blockchain
Answer: b) Hinders blockchain adoption globally
Explanation: Lack of uniform laws creates legal risks.

32. What is a "51% attack"?

a) When a single user controls over half of mining power
b) More than half network uses blockchain
c) All transactions are instant
d) None of the above

Answer: a) When a single user controls over half of mining power
Explanation: This allows manipulation of blockchain transactions.

33. Immutability challenges include ________.

a) Inability to correct errors once stored
b) Data easily altered
c) Automatic deletion
d) Frequent rollbacks

Answer: a) Inability to correct errors once stored
Explanation: Errors once recorded cannot be changed.

34. A privacy issue in public blockchains is ________.

a) Complete anonymity
b) Exposure of transaction details publicly
c) Data encryption
d) Trusted-only access

Answer: b) Exposure of transaction details publicly
Explanation: Public visibility can compromise privacy.

35. What is a node in blockchain?

a) Centralized server
b) Participant computer maintaining blockchain
c) Programming language
d) Data block
Answer: b) Participant computer maintaining blockchain
Explanation: Nodes store, validate, and propagate blockchain data.

36. Which cryptographic function identifies blocks uniquely?

a) Hash function
b) Digital signature
c) Public key
d) Private key
Answer: a) Hash function
Explanation: Hash uniquely identifies each block.

37. What does the consensus mechanism do?

a) Deletes blocks
b) Ensures network agreement
c) Hides transactions
d) Creates private keys

Answer: b) Ensures network agreement
Explanation: Consensus maintains consistency across nodes.

38. Which blockchain is best for enterprise privacy?

a) Public
b) Private
c) Hybrid
d) Consortium
Answer: b) Private
Explanation: Private blockchains restrict access.

39. What is tokenization?

a) Converting assets into digital tokens
b) Hiding identity
c) Destroying records
d) Encrypting personal data
Answer: a) Converting assets into digital tokens
Explanation: Tokenization enables digital asset transfer.

40. Auditability refers to ________.

a) Erasing transactions
b) Traceability and verification
c) Central bank control
d) Private logs
Answer: b) Traceability and verification
Explanation: Blockchain allows full transaction audit trails.

41. Which feature supports financial inclusion?

a) Central authority
b) Open access and no intermediaries
c) High fees
d) Complex licensing
Answer: b) Open access and no intermediaries
Explanation: Blockchain enables access without banks.

42. Smart contracts are based on ________.

a) “if/when…then” logic
b) Manual approval
c) Paper documentation
d) Arbitration

Answer: a) “if/when…then” logic
Explanation: Smart contracts execute automatically on conditions.

43. Which consensus reduces energy compared to PoW?

a) Proof of Work
b) Proof of Stake
c) Proof of Authority
d) Proof of Power
Answer: b) Proof of Stake
Explanation: PoS avoids intensive mining computations.

44. Cross-border payments on blockchain are ________.

a) Slow and costly
b) Faster and cheaper
c) Only for crypto
d) Not traceable
Answer: b) Faster and cheaper
Explanation: Blockchain removes intermediaries.

45. Blockchain transparency affects users by ________.

a) Increasing trust
b) Hiding transactions
c) Removing privacy
d) Centralizing control
Answer: a) Increasing trust
Explanation: Transparency builds confidence in data.

46. Ripple is known for ________.

a) Public blockchain
b) Consortium blockchain
c) Private blockchain
d) Hybrid blockchain

Answer: b) Consortium blockchain
Explanation: Ripple is governed by multiple institutions.

47. When a new node joins a blockchain ________.

a) It receives full blockchain copy
b) Gets limited data
c) Controls network
d) Manages rewards

Answer: a) It receives full blockchain copy
Explanation: Nodes synchronize full ledger.

48. Decentralization enhances security by ________.

a) Making tampering easy
b) Increasing failure risk
c) Multiple copies of data
d) Central server dependency

Answer: c) Multiple copies of data
Explanation: Multiple copies prevent single-point attacks.

49. Gas fees in Ethereum are used to ________.

a) Pay transaction processing
b) Control access
c) Reward tokens
d) None

Answer: a) Pay transaction processing
Explanation: Gas fees compensate miners/validators.

50. Which blockchain pioneered smart contracts?

a) Bitcoin
b) Ethereum
c) Litecoin
d) Dogecoin

Answer: b) Ethereum
Explanation: Ethereum introduced programmable smart contracts.

51. Blockchain is a:
    a) Centralized database
    b) Distributed digital ledger
    c) Spreadsheet
    d) Operating system
    Answer: b) Distributed digital ledger
52. Data in blockchain is stored in the form of:
    a) Tables
    b) Files
    c) Blocks
    d) Sheets
    Answer: c) Blocks
53. Blocks are connected using:
    a) Cables
    b) IP address
    c) Cryptographic hash
    d) HTTP
    Answer: c) Cryptographic hash
54. Once data is stored in blockchain, it:
    a) Can be modified
    b) Can be deleted
    c) Cannot be changed
    d) Can be edited anytime
    Answer: c) Cannot be changed
55. Blockchain follows which model?
    a) Client-server
    b) Peer-to-peer
    c) Centralized
    d) Star topology
    Answer: b) Peer-to-peer
56. Blockchain ensures:
    a) Transparency
    b) Security
    c) Trust
    d) All of the above
    Answer: d) All of the above
57. Traditional systems depend on:
    a) Decentralization
    b) Central authority
    c) Blockchain

d) Peer-to-peer
Answer: b) Central authority

58. Single point of failure exists in:
a) Blockchain
b) Traditional system
c) Peer network
d) Consortium
Answer: b) Traditional system

59. Blockchain removes:
a) Nodes
b) Intermediaries
c) Users
d) Miners
Answer: b) Intermediaries

60. Distributed ledger means:
a) One server
b) Multiple copies of ledger
c) Offline storage
d) Paper record
Answer: b) Multiple copies of ledger

61. Immutability means:
a) Data is editable
b) Data cannot be changed
c) Data deleted
d) Data hidden
Answer: b) Data cannot be changed

62. Blockchain security uses:
a) Cryptographic hashing
b) Digital signatures
c) Encryption
d) All of the above
Answer: d) All of the above

63. Transparency in blockchain means:
a) Hidden transactions
b) Visible transactions
c) No records
d) Manual records
Answer: b) Visible transactions

64. Blockchain is:
a) Centralized
b) Decentralized

c) Offline
d) Isolated
Answer: b) Decentralized

65. A node in blockchain is:
a) Printer
b) Computer/user in network
c) Cable
d) Database
Answer: b) Computer/user in network

66. Miners perform:
a) Cooking
b) Verification process
c) Editing data
d) Deleting blocks
Answer: b) Verification process

67. Consensus protocol is:
a) File format
b) Set of rules
c) Cable type
d) Hash value
Answer: b) Set of rules

68. Each new transaction results in:
a) File deletion
b) New block creation
c) Cable change
d) Printer update
Answer: b) New block creation

69. Blockchain provides fault tolerance because:
a) Single server
b) Multiple nodes
c) Offline system
d) Manual record
Answer: b) Multiple nodes

70. Trustless system means:
a) No trust
b) Third party needed
c) No central authority required
d) Manual approval
Answer: c) No central authority required

71. In traditional system, verification is:
a) Consensus based

b) Third-party based
c) Automatic
d) Blockchain based
Answer: b) Third-party based

72. Blockchain verification is:
a) Manual
b) Consensus based
c) Paper based
d) Printer based
Answer: b) Consensus based

73. Blockchain transaction cost is:
a) High
b) Medium
c) Low
d) Unlimited
Answer: c) Low

74. Data modification in blockchain is:
a) Possible
b) Easy
c) Not possible
d) Allowed
Answer: c) Not possible

75. Public blockchain is:
a) Permissioned
b) Open
c) Private
d) Closed
Answer: b) Open

76. Example of public blockchain:
a) Hyperledger
b) R3 Corda
c) Bitcoin
d) Private chain
Answer: c) Bitcoin

77. Public blockchain disadvantage:
a) High speed
b) High energy consumption
c) Low transparency
d) No scalability
Answer: b) High energy consumption

78. Private blockchain is:
    a) Open
    b) Permissionless
    c) Restricted
    d) Public
    Answer: c) Restricted
79. Private blockchain is controlled by:
    a) Many users
    b) Single organization
    c) Government only
    d) Miners only
    Answer: b) Single organization
80. Consortium blockchain is controlled by:
    a) One person
    b) Group of organizations
    c) Public
    d) None
    Answer: b) Group of organizations
81. Hybrid blockchain combines:
    a) Public and private
    b) Only public
    c) Only private
    d) Only consortium
    Answer: a) Public and private
82. Smart contracts are:
    a) Manual agreements
    b) Self-executing programs
    c) Paper contracts
    d) Email contracts
    Answer: b) Self-executing programs
83. Smart contracts execute when:
    a) Conditions met
    b) Printer connected
    c) Cable attached
    d) Manual approval
    Answer: a) Conditions met
84. Cryptocurrency is:
    a) Physical currency
    b) Digital currency
    c) Paper currency

d) Coin only
Answer: b) Digital currency

85. Cryptocurrency uses:
a) Cryptography
b) Manual approval
c) Printer
d) Fax
Answer: a) Cryptography

86. Bitcoin is example of:
a) Smart contract
b) Cryptocurrency
c) Node
d) Block
Answer: b) Cryptocurrency

87. Blockchain in banking helps:
a) Fraud
b) Secure transfer
c) Delay
d) Manual work
Answer: b) Secure transfer

88. Blockchain in supply chain helps:
a) Counterfeit goods
b) Tracking products
c) Data loss
d) Delay
Answer: b) Tracking products

89. Blockchain in healthcare stores:
a) Patient records
b) TV shows
c) Music
d) Games
Answer: a) Patient records

90. Blockchain voting prevents:
a) Transparency
b) Vote tampering
c) Security
d) Verification
Answer: b) Vote tampering

91. Blockchain in education helps:
a) Certificate verification
b) Gaming

c) Printing
d) Music
Answer: a) Certificate verification

92. Advantage of blockchain:
a) High security
b) Data tampering
c) Manual verification
d) High failure
Answer: a) High security

93. Scalability issue means:
a) Faster processing
b) Slower processing with growth
c) No blocks
d) No data
Answer: b) Slower processing with growth

94. Proof of Work consumes:
a) Low energy
b) High energy
c) No energy
d) Solar only
Answer: b) High energy

95. 51% attack affects:
a) Printer
b) Blockchain security
c) Cloud
d) Database
Answer: b) Blockchain security

96. Storage requirement increases because:
a) Ledger grows
b) Data deletes
c) Nodes reduce
d) Blocks shrink
Answer: a) Ledger grows

97. Blockchain complexity requires:
a) Skilled professionals
b) No knowledge
c) Manual labor
d) Printer
Answer: a) Skilled professionals

98. Regulatory issues arise because:
a) No uniform laws

b) Too many printers
c) Low data
d) Offline system
Answer: a) No uniform laws

99. Privacy challenge arises due to:
a) Transparency
b) Security
c) Encryption
d) Nodes
Answer: a) Transparency

100. Smart contract error leads to:
a) Easy correction
b) Financial loss
c) Data deletion
d) Speed increase
Answer: b) Financial loss

101. Blockchain ledger is stored:
a) In one central server
b) In distributed nodes
c) In printer
d) In offline file
Answer: b) In distributed nodes

102. Cryptographic hash ensures:
a) Data linking
b) Data deletion
c) Data printing
d) Data formatting
Answer: a) Data linking

103. If one block is modified, it:
a) Has no effect
b) Breaks the chain
c) Updates automatically
d) Deletes previous block
Answer: b) Breaks the chain

104. Blockchain improves trust because it is:
a) Manual
b) Transparent
c) Private only
d) Offline
Answer: b) Transparent

105.The smallest building block in blockchain is:
a) Node
b) Block
c) Transaction
d) Miner
Answer: c) Transaction

106.Chain in blockchain means:
a) Cables
b) Sequence of blocks
c) User connection
d) Database
Answer: b) Sequence of blocks

107.Consensus ensures:
a) Agreement among nodes
b) Single authority control
c) Manual approval
d) Printer access
Answer: a) Agreement among nodes

108.Digital signature ensures:
a) Data authenticity
b) Cable safety
c) Printing
d) Formatting
Answer: a) Data authenticity

109.Blockchain reduces fraud because:
a) Data is centralized
b) Data is immutable
c) Data is hidden
d) Data is temporary
Answer: b) Data is immutable

110.Blockchain provides availability due to:
a) Single server
b) Multiple copies
c) Offline storage
d) Printer
Answer: b) Multiple copies

111.Public blockchain allows:
a) Anyone to participate
b) Only admins
c) Only government

d) No users
Answer: a) Anyone to participate

112.Ethereum is example of:
a) Private blockchain
b) Public blockchain
c) Hybrid
d) Consortium
Answer: b) Public blockchain

113.Private blockchain provides:
a) High privacy
b) Public transparency
c) Unlimited access
d) No control
Answer: a) High privacy

114.Consortium blockchain is suitable for:
a) Single company
b) Multiple organizations
c) Public only
d) Printer only
Answer: b) Multiple organizations

115.Hybrid blockchain gives:
a) No flexibility
b) Full openness
c) Controlled transparency
d) No security
Answer: c) Controlled transparency

116.Blockchain speed in public chain is:
a) Faster than private
b) Slower
c) Same as private
d) Unlimited
Answer: b) Slower

117.High energy usage is major issue in:
a) Proof of Work
b) Proof of Stake
c) Private blockchain
d) Consortium
Answer: a) Proof of Work

118.Smart contracts remove need for:
a) Intermediaries
b) Nodes

c) Blocks
d) Miners
Answer: a) Intermediaries

119.Smart contracts are stored on:
a) Printer
b) Blockchain
c) Database
d) Hard disk
Answer: b) Blockchain

120.Cryptocurrency operates on:
a) Blockchain
b) Printer
c) Paper
d) Offline server
Answer: a) Blockchain

121.Blockchain transparency allows:
a) Public verification
b) Hidden data
c) Data deletion
d) Data modification
Answer: a) Public verification

122.Traditional systems are vulnerable due to:
a) Centralization
b) Decentralization
c) Transparency
d) Immutability
Answer: a) Centralization

123.Blockchain fault tolerance exists because:
a) Single server
b) Distributed nodes
c) Manual verification
d) Paper records
Answer: b) Distributed nodes

124.Transaction verification in blockchain is done by:
a) Printer
b) Miners
c) Cable
d) Paper
Answer: b) Miners

125.Consensus mechanism prevents:
a) Fraudulent transactions

b) Transparency
c) Security
d) Blocks
Answer: a) Fraudulent transactions

126.Blockchain ensures data integrity through:
a) Hashing
b) Printing
c) Paper record
d) Formatting
Answer: a) Hashing

127.Peer-to-peer means:
a) Central control
b) Direct user interaction
c) Manual control
d) Offline
Answer: b) Direct user interaction

128.Data tampering is prevented by:
a) Immutability
b) Central authority
c) Manual update
d) Paper
Answer: a) Immutability

129.Blockchain reduces cost by removing:
a) Users
b) Intermediaries
c) Nodes
d) Blocks
Answer: b) Intermediaries

130.Smart contracts execute:
a) Automatically
b) Manually
c) Slowly
d) Offline
Answer: a) Automatically

131.Blockchain improves supply chain by:
a) Tracking goods
b) Deleting records
c) Manual logging
d) Paper
Answer: a) Tracking goods

132.Blockchain in voting ensures:
- a) Transparency
- b) Tampering
- c) Delay
- d) Manual counting

Answer: a) Transparency

133.Blockchain architecture includes:
- a) Nodes
- b) Blocks
- c) Transactions
- d) All of the above

Answer: d) All of the above

134.A block contains:
- a) Set of transactions
- b) Printer data
- c) Cable
- d) IP address

Answer: a) Set of transactions

135.Each block contains previous block's:
- a) Address
- b) Hash
- c) Cable
- d) Signature only

Answer: b) Hash

136.Blockchain storage requirement increases with:
- a) More blocks
- b) Fewer nodes
- c) Printer
- d) Cable

Answer: a) More blocks

137.51% attack occurs when:
- a) Majority control network
- b) Printer fails
- c) Node deletes
- d) Cable breaks

Answer: a) Majority control network

138.Blockchain adoption challenge:
- a) Technical complexity
- b) Manual system
- c) Paper storage

d) Printer
Answer: a) Technical complexity

139.Legal uncertainty affects:
a) Blockchain deployment
b) Cable
c) Printer
d) Router
Answer: a) Blockchain deployment

140.Blockchain privacy issue arises due to:
a) Public visibility
b) Security
c) Encryption
d) Nodes
Answer: a) Public visibility

141.Distributed ledger increases:
a) Availability
b) Vulnerability
c) Deletion
d) Printer speed
Answer: a) Availability

142.Smart contract vulnerability can lead to:
a) Financial loss
b) Printing error
c) Cable damage
d) Light failure
Answer: a) Financial loss

143.Blockchain is suitable for:
a) Secure transactions
b) Paper filing
c) Manual control
d) Offline systems
Answer: a) Secure transactions

144.Blockchain provides higher:
a) Transparency
b) Isolation
c) Delay
d) Manual work
Answer: a) Transparency

145.Blockchain eliminates need for:
a) Trusted third party
b) Nodes

c) Blocks
d) Transactions
Answer: a) Trusted third party

146.Blockchain security is based on:
a) Cryptography
b) Printer
c) Cable
d) Paper
Answer: a) Cryptography

147.Blockchain speed compared to traditional:
a) Always faster
b) Slower in some cases
c) Same
d) No difference
Answer: b) Slower in some cases

148.Proof of Work mainly consumes:
a) Electricity
b) Paper
c) Ink
d) Cable
Answer: a) Electricity

149.Blockchain enables:
a) Decentralized applications
b) Central servers
c) Paper logs
d) Manual verification
Answer: a) Decentralized applications

150.The core idea of blockchain is:
a) Decentralized trust
b) Central authority
c) Printer control
d) Offline record
Answer: a) Decentralized trust

151.Blockchain mainly provides:
a) Centralized control
b) Distributed control
c) Manual control
d) Printer control
Answer: b) Distributed control

152.In blockchain, each block contains:
a) Previous block hash

b) Transaction data
c) Timestamp
d) All of the above
Answer: d) All of the above

153.Timestamp in block is used to:
a) Record transaction time
b) Delete data
c) Print data
d) Encrypt cable
Answer: a) Record transaction time

154.A hash function produces:
a) Fixed length output
b) Variable length output
c) Random cable
d) Printer signal
Answer: a) Fixed length output

155.If any data inside block changes:
a) Hash remains same
b) Hash changes
c) Nothing happens
d) Block deletes
Answer: b) Hash changes

156.Blockchain verification is done before:
a) Block deletion
b) Block addition
c) Printing
d) Storage removal
Answer: b) Block addition

157.Fault tolerance in blockchain means:
a) Network survives node failure
b) Entire network stops
c) Printer fails
d) Block deleted
Answer: a) Network survives node failure

158.Blockchain ensures data consistency because:
a) All nodes have same copy
b) Only one copy exists
c) Data stored offline
d) Data printed
Answer: a) All nodes have same copy

159.Blockchain adoption in finance reduces:
a) Fraud
b) Transparency
c) Security
d) Data
Answer: a) Fraud

160.Public blockchain example includes:
a) Bitcoin
b) Hyperledger
c) R3 Corda
d) Private chain
Answer: a) Bitcoin

161.Private blockchain improves:
a) Privacy
b) Public access
c) Open participation
d) Transparency
Answer: a) Privacy

162.Consortium blockchain is suitable for:
a) Banking groups
b) Individual user only
c) Printer network
d) Cable network
Answer: a) Banking groups

163.Hybrid blockchain allows:
a) Partial transparency
b) No control
c) Full openness only
d) No privacy
Answer: a) Partial transparency

164.Proof of Work is used for:
a) Block validation
b) Data printing
c) Cable encryption
d) Node deletion
Answer: a) Block validation

165.High energy consumption is associated with:
a) Proof of Work
b) Private blockchain
c) Consortium blockchain

d) Hybrid blockchain
Answer: a) Proof of Work

166.Smart contracts reduce:
a) Human error
b) Automation
c) Security
d) Transparency
Answer: a) Human error

167.Smart contracts are coded using:
a) Programming languages
b) Printer language
c) Cable format
d) Paper format
Answer: a) Programming languages

168.Cryptocurrency transactions are:
a) Peer-to-peer
b) Manual
c) Offline
d) Paper-based
Answer: a) Peer-to-peer

169.Blockchain in healthcare ensures:
a) Secure data sharing
b) Data deletion
c) Paper records
d) Manual logs
Answer: a) Secure data sharing

170.Blockchain in supply chain prevents:
a) Counterfeit goods
b) Tracking
c) Transparency
d) Security
Answer: a) Counterfeit goods

171.Blockchain in voting ensures:
a) Tamper-proof voting
b) Manual counting
c) Paper ballots
d) Offline voting
Answer: a) Tamper-proof voting

172.Blockchain improves education by:
a) Secure certificate verification
b) Printing marksheets

c) Deleting records
d) Offline records
Answer: a) Secure certificate verification

173. Blockchain scalability issue refers to:
a) Slow processing at high load
b) Faster processing
c) No transactions
d) Printer failure
Answer: a) Slow processing at high load

174. Storage issue arises because:
a) Ledger grows continuously
b) Blocks shrink
c) Nodes delete data
d) Data removed
Answer: a) Ledger grows continuously

175. 51% attack occurs when:
a) Majority control network
b) Single printer fails
c) Node disconnects
d) Data deleted
Answer: a) Majority control network

176. Private key theft leads to:
a) Security breach
b) Faster speed
c) Data deletion
d) Printer error
Answer: a) Security breach

177. Blockchain complexity limits:
a) Adoption
b) Security
c) Transparency
d) Nodes
Answer: a) Adoption

178. Regulatory issues cause:
a) Legal uncertainty
b) High speed
c) Lower cost
d) Faster adoption
Answer: a) Legal uncertainty

179. Blockchain transparency may cause:
a) Privacy concerns

b) Higher security
c) Faster speed
d) No issues
Answer: a) Privacy concerns

180.Integrating blockchain with traditional systems is:
a) Easy
b) Complex
c) Automatic
d) Free
Answer: b) Complex

181.Initial implementation cost of blockchain is:
a) Low
b) Medium
c) High
d) Zero
Answer: c) High

182.Blockchain adoption is limited due to:
a) Lack of awareness
b) High transparency
c) Low security
d) Immutability
Answer: a) Lack of awareness

183.Smart contract bug can cause:
a) Financial loss
b) Faster speed
c) More nodes
d) Less storage
Answer: a) Financial loss

184.Blockchain improves trust by:
a) Decentralization
b) Central authority
c) Manual approval
d) Paper
Answer: a) Decentralization

185.Distributed ledger increases:
a) Reliability
b) Vulnerability
c) Deletion
d) Failure
Answer: a) Reliability

186.Blockchain data once added becomes:
a) Mutable
b) Immutable
c) Temporary
d) Editable
Answer: b) Immutable

187.Peer-to-peer network means:
a) Direct node interaction
b) Central server
c) Manual control
d) Offline system
Answer: a) Direct node interaction

188.Blockchain is suitable for:
a) Secure digital transactions
b) Manual paper work
c) Printer networks
d) Offline systems
Answer: a) Secure digital transactions

189.Hash linking ensures:
a) Data integrity
b) Data deletion
c) Cable safety
d) Printer output
Answer: a) Data integrity

190.Blockchain consensus ensures:
a) Network agreement
b) Central authority
c) Printer
d) Cable
Answer: a) Network agreement

191.Digital signatures ensure:
a) Authentication
b) Data deletion
c) Printer error
d) Speed
Answer: a) Authentication

192.Blockchain increases transparency by:
a) Allowing transaction verification
b) Hiding records
c) Deleting data

d) Manual approval
Answer: a) Allowing transaction verification

193.Blockchain removes single point of failure because:
a) Distributed network
b) Single server
c) Printer
d) Paper
Answer: a) Distributed network

194.Cryptocurrency example includes:
a) Bitcoin
b) Printer
c) Router
d) Cable
Answer: a) Bitcoin

195.Blockchain reduces transaction time by:
a) Removing intermediaries
b) Adding manual steps
c) Printing
d) Offline storage
Answer: a) Removing intermediaries

196.Blockchain architecture mainly includes:
a) Nodes
b) Blocks
c) Transactions
d) All of the above
Answer: d) All of the above

197.Mining is process of:
a) Verifying transactions
b) Deleting blocks
c) Printing
d) Cable installation
Answer: a) Verifying transactions

198.Blockchain security is based on:
a) Cryptography
b) Paper
c) Printer
d) Cable
Answer: a) Cryptography

199.Blockchain provides high:
a) Security
b) Transparency

c) Reliability
d) All of the above
Answer: d) All of the above

200.Core principle of blockchain technology is:
a) Decentralized trust system
b) Centralized control
c) Manual approval
d) Printer network
Answer: a) Decentralized trust system

UNIT 4 – IMMERSIVE TECHNOLOGY AND SUSTAINABLE COMPUTING

1. Immersive technology mainly creates:
   a) Paper records
   b) Digital interactive environments
   c) Cable systems
   d) Offline files
   Answer: b) Digital interactive environments
2. Immersive systems enhance which type of experience?
   a) Sensory
   b) Manual
   c) Printed
   d) Offline
   Answer: a) Sensory
3. VR creates a:
   a) Partial environment
   b) Fully artificial environment
   c) Real-only environment
   d) Paper-based system
   Answer: b) Fully artificial environment
4. VR users cannot see:
   a) Digital objects
   b) Real world
   c) Audio
   d) Motion
   Answer: b) Real world
5. VR requires:
   a) Keyboard only
   b) VR headset
   c) Printer
   d) Fax
   Answer: b) VR headset
6. VR is widely used in:
   a) Flight simulation
   b) Paper printing
   c) Radio
   d) Telephone
   Answer: a) Flight simulation
7. AR enhances:
   a) Only digital world
   b) Real world with digital content

c) Offline world
d) Paper records
Answer: b) Real world with digital content

8. AR works mainly through:
a) Smartphones
b) Fax
c) Printer
d) Typewriter
Answer: a) Smartphones

9. AR does NOT:
a) Replace real world
b) Add digital objects
c) Provide interaction
d) Enhance environment
Answer: a) Replace real world

10. Pokémon GO is example of:
a) VR
b) AR
c) MR
d) XR
Answer: b) AR

11. MR combines:
a) Only VR
b) Only AR
c) VR and AR
d) None
Answer: c) VR and AR

12. MR allows digital objects to:
a) Remain static
b) Interact with real objects
c) Be invisible
d) Delete data
Answer: b) Interact with real objects

13. XR stands for:
a) Extended Reality
b) External Reality
c) Extra Rendering
d) Extended Robotics
Answer: a) Extended Reality

14. XR includes:
a) VR
b) AR
c) MR

d) All of the above
Answer: d) All of the above

15. Haptic technology provides:
    a) Sound feedback
    b) Touch feedback
    c) Visual feedback only
    d) Data storage
    Answer: b) Touch feedback
16. Haptic devices simulate:
    a) Temperature
    b) Touch sensation
    c) Smell
    d) Color
    Answer: b) Touch sensation
17. VR improves understanding through:
    a) Simulation
    b) Paper
    c) Manual
    d) Cable
    Answer: a) Simulation
18. AR improves:
    a) Real-time information
    b) Offline data
    c) Printer speed
    d) Cable length
    Answer: a) Real-time information
19. MR is used in:
    a) Remote collaboration
    b) Fax
    c) Printer
    d) Paper
    Answer: a) Remote collaboration
20. Immersive tech is used in:
    a) Healthcare
    b) Education
    c) Gaming
    d) All of the above
    Answer: d) All of the above
21. VR in healthcare is mainly used for:
    a) Surgery simulation
    b) Printing
    c) Radio

d) Fax
Answer: a) Surgery simulation

22. VR helps in dangerous training because it is:
a) Risk-free
b) Slow
c) Manual
d) Offline
Answer: a) Risk-free

23. AR navigation displays:
a) Digital directions
b) Paper map
c) Cable
d) Printer
Answer: a) Digital directions

24. Immersive tech improves:
a) Industrial safety
b) Manual errors
c) Delay
d) Paper logs
Answer: a) Industrial safety

25. VR tourism allows:
a) Virtual tours
b) Physical travel
c) Cable
d) Printer
Answer: a) Virtual tours

26. MR uses:
a) Sensors and cameras
b) Paper
c) Ink
d) Fax
Answer: a) Sensors and cameras

27. VR gaming requires:
a) Immersive headset
b) Typewriter
c) Printer
d) Router
Answer: a) Immersive headset

28. AR product visualization helps in:
a) Marketing
b) Cable repair
c) Printing

d) Paper filing
Answer: a) Marketing

29. XR offers:
a) Flexible immersive solutions
b) Manual records
c) Paper
d) Fax
Answer: a) Flexible immersive solutions

30. Haptic feedback enhances:
a) Realism
b) Paper
c) Cable
d) Printer
Answer: a) Realism

31. Green computing aims to reduce:
a) Energy consumption
b) Data
c) Users
d) Servers
Answer: a) Energy consumption

32. Green IT promotes:
a) Sustainability
b) Pollution
c) Paper waste
d) Data loss
Answer: a) Sustainability

33. Green computing reduces:
a) Carbon emission
b) Storage
c) Network
d) Devices
Answer: a) Carbon emission

34. Virtualization reduces:
a) Physical servers
b) Users
c) Network
d) Printers
Answer: a) Physical servers

35. Data centers consume:
a) Large electricity
b) Low electricity
c) No electricity

d) Solar only
Answer: a) Large electricity

36. Efficient cooling helps:
a) Save energy
b) Increase heat
c) Slow server
d) Delete data
Answer: a) Save energy

37. E-waste includes:
a) Discarded electronics
b) Paper
c) Ink
d) Water
Answer: a) Discarded electronics

38. Improper e-waste disposal causes:
a) Pollution
b) Speed
c) Transparency
d) Security
Answer: a) Pollution

39. Recycling recovers:
a) Valuable materials
b) Paper
c) Cable
d) Fax
Answer: a) Valuable materials

40. First step in e-waste recycling:
a) Collection
b) Printing
c) Burning
d) Deleting
Answer: a) Collection

41. Magnetic separation is used for:
a) Metal separation
b) Paper
c) Plastic
d) Water
Answer: a) Metal separation

42. Eddy current separation removes:
a) Non-ferrous metals
b) Paper
c) Plastic

d) Ink
Answer: a) Non-ferrous metals

43. Green computing lowers:
    a) Operational cost
    b) Productivity
    c) Security
    d) Transparency
    Answer: a) Operational cost
44. Water separation is used in:
    a) Recycling process
    b) VR
    c) AR
    d) MR
    Answer: a) Recycling process
45. Sustainable IT practices help:
    a) Environment
    b) Pollution
    c) Waste
    d) Carbon increase
    Answer: a) Environment
46. Energy-efficient hardware:
    a) Reduces power usage
    b) Increases heat
    c) Slows system
    d) Deletes data
    Answer: a) Reduces power usage
47. Green computing focuses on:
    a) Eco-friendly computing
    b) Fast gaming
    c) Printing
    d) Fax
    Answer: a) Eco-friendly computing
48. Data center optimization improves:
    a) Efficiency
    b) Pollution
    c) Delay
    d) Paper waste
    Answer: a) Efficiency
49. Green IT encourages:
    a) Recycling
    b) Dumping
    c) Burning

d) Deleting
Answer: a) Recycling

50. Proper e-waste management prevents:
 a) Environmental hazards
 b) Transparency
 c) Security
 d) Printing
 Answer: a) Environmental hazards

51. Quantum computing uses:
 a) Quantum mechanics
 b) Printer
 c) Paper
 d) Cable
 Answer: a) Quantum mechanics

52. Basic unit of quantum computing:
 a) Bit
 b) Qubit
 c) Byte
 d) Block
 Answer: b) Qubit

53. Qubit can exist in:
 a) 0 only
 b) 1 only
 c) Both 0 and 1
 d) None
 Answer: c) Both 0 and 1

54. Superposition allows:
 a) Parallel processing
 b) Sequential only
 c) Printing
 d) Cable
 Answer: a) Parallel processing

55. Entanglement links:
 a) Two qubits
 b) Two printers
 c) Two cables
 d) Two routers
 Answer: a) Two qubits

56. Quantum interference helps:
 a) Amplify correct results
 b) Delete blocks
 c) Print faster

d) Cable repair
Answer: a) Amplify correct results

57. Classical computer uses:
a) Bits
b) Qubits
c) Blocks
d) Hash
Answer: a) Bits

58. Quantum computer supports:
a) Parallel processing
b) Manual processing
c) Offline processing
d) Paper processing
Answer: a) Parallel processing

59. IBM Quantum is example of:
a) Classical computer
b) Quantum computer
c) Printer
d) Router
Answer: b) Quantum computer

60. Quantum computing helps in:
a) Cryptography
b) Drug discovery
c) Optimization
d) All of the above
Answer: d) All of the above

61. Quantum computers may break:
a) RSA encryption
b) Printer
c) Cable
d) Fax
Answer: a) RSA encryption

62. Quantum computers simulate:
a) Molecular structures
b) Paper
c) Printer
d) Cable
Answer: a) Molecular structures

63. Qubit stability is:
a) Challenge
b) Advantage
c) Printer

d) Cable
Answer: a) Challenge

64. Quantum error correction is:
a) Difficult
b) Easy
c) Printer-based
d) Cable-based
Answer: a) Difficult

65. Quantum computing cost is:
a) High
b) Low
c) Free
d) Zero
Answer: a) High

66. Quantum computing enhances:
a) AI
b) Printing
c) Cable
d) Fax
Answer: a) AI

67. Exponential growth in power depends on:
a) Number of qubits
b) Printer
c) Cable
d) Router
Answer: a) Number of qubits

68. Quantum systems are still:
a) Developing
b) Complete
c) Finished
d) Manual
Answer: a) Developing

69. Quantum computing is best for:
a) Complex problems
b) Simple typing
c) Printing
d) Paper filing
Answer: a) Complex problems

70. Future of quantum computing may:
a) Revolutionize computing
b) Stop computing
c) Replace paper

d) Delete internet
Answer: a) Revolutionize computing

116. Virtual Reality system mainly consists of:

a) Head-mounted display
b) Sensors
c) Controllers
d) All of the above
Answer: d) All of the above

117. Field of View (FOV) in VR affects:

a) Immersion level
b) Printer speed
c) Cable length
d) Storage size
Answer: a) Immersion level

118. Motion tracking in VR is used to:

a) Detect user movement
b) Delete files
c) Print faster
d) Store data
Answer: a) Detect user movement

119. AR markers are used to:

a) Align digital objects
b) Print documents
c) Connect cables
d) Delete records
Answer: a) Align digital objects

120. Markerless AR uses:

a) GPS and sensors
b) Paper codes
c) Printer ink
d) Fax signals
Answer: a) GPS and sensors

121. Immersive tech in education increases:

a) Student engagement
b) Paper usage
c) Manual learning
d) Delay
Answer: a) Student engagement

122. VR sickness is caused due to:

a) Motion mismatch
b) Cable fault
c) Printer issue
d) Storage error
Answer: a) Motion mismatch

123. AR is useful in maintenance because it:

a) Displays repair instructions
b) Deletes files
c) Prints manuals
d) Disconnects network
Answer: a) Displays repair instructions

124. Mixed Reality devices include:

a) Microsoft HoloLens
b) Printer
c) Router
d) Keyboard
Answer: a) Microsoft HoloLens

125. XR technology improves:

a) Remote collaboration
b) Paper filing
c) Manual entry
d) Cable testing
Answer: a) Remote collaboration

126. Power Usage Effectiveness (PUE) measures:

a) Data center efficiency
b) Printer speed
c) Cable length
d) Storage capacity
Answer: a) Data center efficiency

127. Green cloud computing reduces:

a) Carbon footprint
b) Data
c) Nodes
d) Routers
Answer: a) Carbon footprint

128. Sustainable IT procurement encourages:

a) Eco-friendly hardware
b) High power devices
c) Manual systems

d) Paper systems
Answer: a) Eco-friendly hardware

129. E-waste recycling helps recover:

a) Gold
b) Silver
c) Copper
d) All of the above
Answer: d) All of the above

130. Biodegradable components reduce:

a) Environmental impact
b) Network speed
c) Data storage
d) Printer usage
Answer: a) Environmental impact

131. Server virtualization allows:

a) Multiple OS on one machine
b) One OS per server
c) Printer control
d) Cable removal
Answer: a) Multiple OS on one machine

132. Sleep mode in devices helps:

a) Save energy
b) Increase power use
c) Delete files
d) Print faster
Answer: a) Save energy

133. Renewable energy in data centers uses:

a) Solar power
b) Wind power
c) Hydro power
d) All of the above
Answer: d) All of the above

134. Proper disposal of batteries prevents:

a) Toxic leakage
b) Faster printing
c) Cable damage
d) Storage issues
Answer: a) Toxic leakage

135. Smart grids improve:

a) Energy distribution efficiency
b) Paper work
c) Printing
d) Manual logging
Answer: a) Energy distribution efficiency

136. Quantum annealing is mainly used for:

a) Optimization problems
b) Printing
c) Data deletion
d) Cable management
Answer: a) Optimization problems

137. Shor's algorithm is used for:

a) Factoring large numbers
b) Printing
c) File compression
d) Data deletion
Answer: a) Factoring large numbers

138. Grover's algorithm improves:

a) Search speed
b) Printing
c) Cable testing
d) Typing speed
Answer: a) Search speed

139. Quantum cryptography provides:

a) Secure key distribution
b) Printer encryption
c) Cable testing
d) Offline security
Answer: a) Secure key distribution

140. Quantum teleportation transfers:

a) Quantum state
b) Printer
c) Cable
d) Hardware
Answer: a) Quantum state

141. Decoherence limits:

a) Computation accuracy
b) Printer output
c) Storage

d) Network
Answer: a) Computation accuracy

142. Quantum supremacy demonstrates:

a) Quantum advantage over classical
b) Printer speed
c) Cable efficiency
d) Paper system
Answer: a) Quantum advantage over classical

143. Qubit coherence time affects:

a) Stability
b) Printer
c) Cable
d) Paper
Answer: a) Stability

144. Superconducting qubits operate at:

a) Very low temperatures
b) Room temperature
c) High temperature
d) Outdoor temperature
Answer: a) Very low temperatures

145. Quantum computers are best suited for:

a) Complex simulations
b) Basic typing
c) Printing
d) Fax
Answer: a) Complex simulations

146. Classical computing is based on:

a) Boolean logic
b) Quantum physics
c) Superposition
d) Entanglement
Answer: a) Boolean logic

147. Quantum computing research is led by:

a) IBM
b) Google
c) Microsoft
d) All of the above
Answer: d) All of the above

148. Quantum computing may impact cybersecurity by:

a) Breaking traditional encryption
b) Printing faster
c) Cable testing
d) Paper removal
Answer: a) Breaking traditional encryption

149. One major barrier in quantum computing is:

a) Hardware complexity
b) Paper
c) Printer
d) Cable
Answer: a) Hardware complexity

150. The future of quantum computing is expected to:

a) Transform computing industry
b) Replace printers
c) Remove cables
d) Eliminate internet
Answer: a) Transform computing industry

UNIT 5 – DIGITAL FORENSICS & ETHICAL HACKING

1. Digital forensics primarily deals with:

a) Physical evidence only
b) Digital evidence investigation
c) Paper records
d) Manual auditing
Answer: b) Digital evidence investigation

2. Digital evidence must be:

a) Modified
b) Deleted
c) Preserved carefully
d) Edited
Answer: c) Preserved carefully

3. A forensic image is:

a) Screenshot
b) Exact bit-by-bit copy
c) Photo of device
d) Backup file
Answer: b) Exact bit-by-bit copy

4. Hash values are used to:

a) Encrypt data
b) Verify integrity
c) Delete files
d) Format disk
Answer: b) Verify integrity

5. Chain of custody ensures:

a) File compression
b) Accountability of evidence handling
c) Data deletion
d) Fast processing
Answer: b) Accountability of evidence handling

6. Volatile data is found in:

a) Hard disk
b) RAM
c) DVD
d) USB only
Answer: b) RAM

7. File carving is used to:

a) Print files
b) Recover deleted files
c) Encrypt files
d) Compress files
Answer: b) Recover deleted files

8. Timeline reconstruction helps in:

a) Speeding system
b) Understanding sequence of events
c) Printing logs
d) Deleting malware
Answer: b) Understanding sequence of events

9. DFRWS model was proposed in:

a) 1998
b) 2001
c) 2005
d) 2010
Answer: b) 2001

10. ADFM adds which extra phase?

a) Presentation
b) Preparation
c) Analysis
d) Examination
Answer: b) Preparation

11. IDIP integrates:

a) Digital & physical crime scenes
b) Printer & network
c) Software & hardware
d) OS & database
Answer: a) Digital & physical crime scenes

12. EEDIP emphasizes:

a) Automation & traceability
b) Paper work
c) Manual logging
d) Printing
Answer: a) Automation & traceability

13. UMDFPM uses:

a) Java
b) UML diagrams
c) Python
d) SQL
Answer: b) UML diagrams

14. Examination stage involves:

a) Filtering relevant data
b) Destroying data
c) Printing data
d) Formatting
Answer: a) Filtering relevant data

15. Preservation stage ensures:

a) Data integrity
b) Data deletion
c) Data duplication
d) Data printing
Answer: a) Data integrity

16. Evidence handling begins with:

a) Presentation
b) Seizure
c) Analysis
d) Documentation
Answer: b) Seizure

17. Forensic tools must be:

a) Random
b) Scientifically accepted
c) Manual
d) Free only
Answer: b) Scientifically accepted

18. Never trust the operating system because:

a) It may be compromised
b) It is slow
c) It deletes files
d) It prints logs
Answer: a) It may be compromised

19. Ethical hacking is:

a) Illegal hacking
b) Authorized security testing
c) Cybercrime
d) Data theft
Answer: b) Authorized security testing

20. Ethical hackers must obtain:

a) Password
b) Legal permission
c) IP address
d) Cable
Answer: b) Legal permission

21. Reconnaissance phase involves:

a) Information gathering
b) Data deletion
c) Printing
d) Encryption
Answer: a) Information gathering

22. Scanning phase identifies:

a) Open ports
b) Printer speed
c) Cable faults
d) Storage
Answer: a) Open ports

23. Exploitation phase attempts:

a) Controlled attack
b) Data deletion
c) Manual login
d) Printing
Answer: a) Controlled attack

24. White hat hackers are:

a) Criminals
b) Authorized professionals
c) Hacktivists
d) Spammers
Answer: b) Authorized professionals

25. Black hat hackers act:

a) Legally
b) Illegally
c) Academically
d) Publicly
Answer: b) Illegally

26. Grey hat hackers:

a) Always legal
b) Always illegal
c) Operate without permission but no malicious intent
d) Work for government only
Answer: c) Operate without permission but no malicious intent

27. Blue hat hackers are usually:

a) Experts
b) Beginners
c) Government officials
d) Judges
Answer: b) Beginners

28. Green hat hackers are:

a) Learning hackers
b) Criminal hackers
c) Bank officials
d) Auditors
Answer: a) Learning hackers

29. Red hat hackers:

a) Fight black hats aggressively
b) Print data
c) Use paper logs
d) Work offline
Answer: a) Fight black hats aggressively

30. Network hacking targets:

a) Routers and switches
b) Paper files
c) Printers
d) Cameras only
Answer: a) Routers and switches

31. AI-powered phishing uses:
    a) Machine learning
    b) Cable
    c) Printer
    d) Router
    Answer: a) Machine learning
32. Ransomware 2.0 includes:
    a) Double extortion
    b) Triple extortion
    c) Data theft
    d) All of the above
    Answer: d) All of the above
33. IoT exploits target:
    a) Smart devices
    b) Paper
    c) Printer
    d) Fax
    Answer: a) Smart devices
34. Deepfake uses:
    a) Machine learning
    b) Cable
    c) Printer
    d) Router
    Answer: a) Machine learning
35. OS downgrade attack installs:
    a) Latest OS

b) Older vulnerable OS
c) Secure OS
d) Updated OS
Answer: b) Older vulnerable OS

36. Firmware exploits target:
a) BIOS/UEFI
b) Printer
c) Cable
d) Mouse
Answer: a) BIOS/UEFI

37. WAF bypass techniques attempt to:
a) Evade firewall filters
b) Delete files
c) Print logs
d) Disconnect router
Answer: a) Evade firewall filters

38. Zero-day vulnerability is:
a) Known and patched
b) Unknown and unpatched
c) Printer error
d) Cable break
Answer: b) Unknown and unpatched

39. NCSP 2013 protects:
a) National cyberspace
b) Printer
c) Cable
d) Paper
Answer: a) National cyberspace

40. IT Act 2000 provides legal recognition to:
a) Electronic records
b) Paper files
c) Fax
d) Printer
Answer: a) Electronic records

41. Section 66C deals with:
a) Identity theft
b) Printer
c) Cable
d) Router
Answer: a) Identity theft

42. Section 66F deals with:
a) Cyber terrorism
b) Paper
c) Printer

d) Cable
Answer: a) Cyber terrorism

43. DPDP Act 2023 protects:
    a) Personal data
    b) Printer
    c) Router
    d) Cable
    Answer: a) Personal data
44. Data Principal refers to:
    a) Individual
    b) Company
    c) Router
    d) Printer
    Answer: a) Individual
45. Data Fiduciary refers to:
    a) Organization handling data
    b) Printer
    c) Cable
    d) Router
    Answer: a) Organization handling data
46. CCPWC scheme protects:
    a) Women & children
    b) Printers
    c) Routers
    d) Servers
    Answer: a) Women & children
47. CERT-In is responsible for:
    a) Cyber incident response
    b) Printing
    c) Cable repair
    d) OS installation
    Answer: a) Cyber incident response
48. Section 69 allows government to:
    a) Intercept data
    b) Print data
    c) Delete logs
    d) Remove OS
    Answer: a) Intercept data
49. Section 43 deals with:
    a) Unauthorized access
    b) Printer
    c) Cable
    d) Mouse
    Answer: a) Unauthorized access

50. Ethical hacking improves:
    a) Security posture
    b) Printer speed
    c) Cable
    d) Paper
    Answer: a) Security posture

51. In digital forensics, identification stage involves:

a) Recovering deleted files
b) Recognizing incident occurrence
c) Reporting findings
d) Court presentation
Answer: b) Recognizing incident occurrence

52. Preservation phase prevents:

a) Evidence tampering
b) Evidence printing
c) Evidence deletion
d) Evidence formatting
Answer: a) Evidence tampering

53. Collection phase gathers:

a) Relevant digital data
b) Paper documents
c) Printer logs only
d) Manual notes
Answer: a) Relevant digital data

54. Examination phase removes:

a) Relevant data
b) Irrelevant data
c) Entire evidence
d) Chain of custody
Answer: b) Irrelevant data

55. Presentation stage requires:

a) Clear documentation
b) Deleting files
c) Formatting drives
d) Printing random logs
Answer: a) Clear documentation

56. Decision phase in DFRWS involves:

a) Legal action
b) Printing reports
c) System shutdown
d) Data deletion
Answer: a) Legal action

57. ADFM Return of Evidence phase ensures:

a) Proper device return
b) Device deletion
c) Printer usage
d) Cable removal
Answer: a) Proper device return

58. IDIP readiness phase ensures:

a) Preparation before incident
b) Deleting malware
c) Printer installation
d) OS upgrade
Answer: a) Preparation before incident

59. Deployment phase begins when:

a) Incident detected
b) Printer installed
c) Cable connected
d) OS updated
Answer: a) Incident detected

60. Review phase in IDIP focuses on:

a) Improving future investigations
b) Deleting evidence
c) Printing documents
d) Formatting disk
Answer: a) Improving future investigations

61. EEDIP Authorization phase ensures:

a) Legal permission
b) Printer control
c) Cable security
d) Router access
Answer: a) Legal permission

62. Extended model supports:

a) Multi-agency coordination
b) Single investigator only
c) Printer network
d) Cable testing
Answer: a) Multi-agency coordination

63. Use case diagram shows:

a) Actors and roles
b) Printer layout
c) Cable path
d) Storage capacity
Answer: a) Actors and roles

64. Activity diagram represents:

a) Workflow
b) File system
c) Network cable
d) Storage format
Answer: a) Workflow

65. Sequence diagram shows:

a) Interaction order
b) Printer speed
c) Cable order
d) Router configuration
Answer: a) Interaction order

66. RAM data should be collected:

a) After shutdown
b) Before shutdown
c) Never
d) After formatting
Answer: b) Before shutdown

67. Steganography detection finds:

a) Hidden data
b) Printed files
c) Deleted cables
d) Offline systems
Answer: a) Hidden data

68. Keyword search helps identify:

a) Suspicious terms
b) Printer speed
c) Cable type
d) OS version
Answer: a) Suspicious terms

69. Documentation stage increases:

a) Legal credibility
b) Printer usage
c) Cable usage
d) OS upgrade
Answer: a) Legal credibility

70. Ethical hackers simulate:

a) Real attacks legally
b) Paper filing
c) Printing
d) Cable testing
Answer: a) Real attacks legally

71. Maintaining access phase checks:

a) Persistence
b) Printing
c) Cable
d) Storage
Answer: a) Persistence

72. Reporting phase includes:

a) Vulnerability details
b) Printer logs only
c) Cable diagram
d) OS update
Answer: a) Vulnerability details

73. Ethical hacking improves:

a) Incident response readiness
b) Paper storage
c) Printer ink
d) Cable repair
Answer: a) Incident response readiness

74. AI phishing emails often show:

a) Perfect grammar
b) Spelling errors only
c) Printer logs
d) Cable details
Answer: a) Perfect grammar

75. Double extortion ransomware includes:

a) Encryption + data theft
b) Encryption only
c) Deletion only
d) Printer attack
Answer: a) Encryption + data theft

76. Triple extortion adds:

a) DDoS or blackmail
b) Printer repair
c) Cable testing
d) OS upgrade
Answer: a) DDoS or blackmail

77. Indicators of ransomware include:

a) Mass file encryption
b) Printer slowdown
c) Cable noise
d) Paper jam
Answer: a) Mass file encryption

78. EDR stands for:

a) Endpoint Detection and Response
b) Electronic Data Recovery
c) Evidence Data Removal
d) Emergency Digital Router
Answer: a) Endpoint Detection and Response

79. IoT default passwords cause:

a) Vulnerability
b) Printer speed
c) Cable issue
d) Storage expansion
Answer: a) Vulnerability

80. VLAN helps in:

a) Network isolation
b) Printer repair
c) Cable deletion
d) File formatting
Answer: a) Network isolation

81. Deepfake detection checks:

a) Metadata
b) Printer logs
c) Cable signals
d) Router configuration
Answer: a) Metadata

82. OS hacking targets:

a) System vulnerabilities
b) Paper files
c) Printer drivers
d) Cable
Answer: a) System vulnerabilities

83. Secure boot prevents:

a) Unauthorized firmware
b) Printer errors
c) Cable faults
d) Storage errors
Answer: a) Unauthorized firmware

84. TPM stands for:

a) Trusted Platform Module
b) Temporary Print Module
c) Technical Printer Model
d) Terminal Port Manager
Answer: a) Trusted Platform Module

85. SQL injection is:

a) Application vulnerability
b) Printer failure
c) Cable break
d) Router issue
Answer: a) Application vulnerability

86. Log4j was example of:

a) Zero-day exploit
b) Printer bug
c) Cable error
d) OS update
Answer: a) Zero-day exploit

87. NCSP focuses on protecting:

a) Critical infrastructure
b) Printer network
c) Cable design
d) Router layout
Answer: a) Critical infrastructure

88. CERT-In stands for:

a) Computer Emergency Response Team – India
b) Central Router Testing
c) Cyber Equipment Repair Team
d) Certified Router Inspector
Answer: a) Computer Emergency Response Team – India

89. Section 66D deals with:

a) Online cheating
b) Printer misuse
c) Cable theft
d) OS downgrade
Answer: a) Online cheating

90. Section 72A relates to:

a) Breach of confidentiality
b) Printer hacking
c) Cable theft
d) OS upgrade
Answer: a) Breach of confidentiality

91. DPDP requires:

a) User consent
b) Printer logs
c) Cable testing
d) Router approval
Answer: a) User consent

92. Data Protection Board oversees:

a) Compliance
b) Printer repair
c) Cable usage
d) Router traffic
Answer: a) Compliance

93. CCPWC portal allows:

a) Cybercrime reporting
b) Printer installation
c) Cable registration
d) OS activation
Answer: a) Cybercrime reporting

94. Cyber stalking is:

a) Online harassment
b) Printer misuse
c) Cable theft
d) Router attack
Answer: a) Online harassment

95. Evidence integrity is verified using:

a) Hash values
b) Printer log
c) Cable ID
d) OS version
Answer: a) Hash values

96. Memory dump analysis helps in:

a) Malware detection
b) Printer repair
c) Cable fixing
d) Paper filing
Answer: a) Malware detection

97. Forensic readiness improves:

a) Investigation efficiency
b) Printer speed
c) Cable length
d) Storage
Answer: a) Investigation efficiency

98. Legal admissibility requires:

a) Proper chain of custody
b) Printer
c) Cable
d) Router
Answer: a) Proper chain of custody

99. Ethical hacking reduces:

a) Security risks
b) Printer errors
c) Cable damage
d) OS bugs
Answer: a) Security risks

100. The ultimate aim of digital forensics is:

a) Court-admissible evidence
b) Printing
c) Cable repair
d) OS update
Answer: a) Court-admissible evidence

101. In digital forensics, integrity ensures that evidence is:

a) Modified
b) Authentic and unchanged
c) Deleted
d) Printed
Answer: b) Authentic and unchanged

102. A write blocker is used to:

a) Prevent data modification
b) Delete files
c) Format drives
d) Encrypt systems
Answer: a) Prevent data modification

103. Forensic duplication is also known as:

a) Logical backup
b) Bit-stream copy
c) Data deletion
d) Compression
Answer: b) Bit-stream copy

104. Volatile evidence includes:

a) Hard disk files
b) RAM contents
c) Archived backups
d) Printed documents
Answer: b) RAM contents

105. Log analysis helps investigators detect:

a) System activities
b) Printer errors
c) Cable failures
d) OS installation
Answer: a) System activities

106. Evidence should be stored in:

a) Open area
b) Secure environment
c) Shared folder
d) Public network
Answer: b) Secure environment

107. MD5 and SHA are examples of:

a) Encryption keys
b) Hash algorithms
c) Firewalls
d) Antivirus tools
Answer: b) Hash algorithms

108. Forensic soundness means:

a) Using approved tools
b) Deleting data
c) Modifying logs
d) Skipping documentation
Answer: a) Using approved tools

109. Chain of custody includes:

a) Who handled evidence
b) When it was handled
c) Why it was handled
d) All of the above
Answer: d) All of the above

110. Preparation phase in investigation ensures:

a) Legal readiness
b) Data deletion
c) Cable setup
d) Printer installation
Answer: a) Legal readiness

111. Ethical hacking primarily identifies:

a) Security vulnerabilities
b) Printer speed
c) Cable length
d) Storage size
Answer: a) Security vulnerabilities

112. Penetration testing is conducted:

a) Without permission
b) With authorization
c) Secretly
d) Illegally
Answer: b) With authorization

113. Vulnerability assessment focuses on:

a) Identifying weaknesses
b) Deleting systems
c) Formatting drives
d) Printing logs
Answer: a) Identifying weaknesses

114. Social engineering attacks exploit:

a) Human psychology
b) Hardware failure
c) Cable damage
d) Printer malfunction
Answer: a) Human psychology

115. Multi-Factor Authentication (MFA) improves:

a) Security
b) Printer performance
c) Cable speed
d) Storage
Answer: a) Security

116. SPF, DKIM, and DMARC protect against:

a) Phishing emails
b) Printer hacks
c) Cable faults
d) OS errors
Answer: a) Phishing emails

117. Double extortion ransomware threatens:

a) Data publication
b) Printer damage
c) Cable theft
d) OS crash
Answer: a) Data publication

118. Endpoint Detection and Response tools monitor:

a) End-user devices
b) Printers only
c) Routers only
d) Cables only
Answer: a) End-user devices

119. Firmware malware can survive:

a) OS reinstallation
b) Cable replacement
c) Printer reset
d) File deletion
Answer: a) OS reinstallation

120. Zero-day exploit targets:

a) Unknown vulnerabilities
b) Patched systems
c) Printer firmware
d) Paper records
Answer: a) Unknown vulnerabilities

121. Section 66B relates to:

a) Receiving stolen computer resources
b) Printer misuse
c) Cable theft
d) OS downgrade
Answer: a) Receiving stolen computer resources

122. Section 66C addresses:

a) Identity theft
b) Printer theft
c) Cable misuse
d) Data compression
Answer: a) Identity theft

123. Section 66D punishes:

a) Cheating by personation
b) Printer hacking
c) Cable damage
d) Storage deletion
Answer: a) Cheating by personation

124. Section 66F defines:

a) Cyber terrorism
b) Printer malfunction
c) Router error
d) Cable break
Answer: a) Cyber terrorism

125. Section 72A deals with:

a) Breach of confidentiality
b) Printer logs
c) Cable ID
d) OS update
Answer: a) Breach of confidentiality

126. DPDP Act 2023 requires organizations to:

a) Obtain user consent
b) Delete user data
c) Print user logs
d) Share data publicly
Answer: a) Obtain user consent

127. Data Principal has right to:

a) Access personal data
b) Delete printer
c) Change cable
d) Update router
Answer: a) Access personal data

128. Data Fiduciary must ensure:

a) Lawful data processing
b) Printer control
c) Cable testing
d) OS upgrade
Answer: a) Lawful data processing

129. CCPWC portal enables:

a) Reporting cybercrime
b) Printing complaints
c) Cable setup
d) Router activation
Answer: a) Reporting cybercrime

130. CERT-In provides:

a) Incident response
b) Printer repair
c) Cable installation
d) Storage
Answer: a) Incident response

131. Forensic imaging tools must:

a) Preserve metadata
b) Delete timestamps
c) Modify files
d) Compress logs
Answer: a) Preserve metadata

132. Metadata analysis helps determine:

a) File creation time
b) Printer speed
c) Cable type
d) Router model
Answer: a) File creation time

133. In ransomware case, volatile memory capture helps find:

a) Encryption keys
b) Printer ID
c) Cable number
d) OS version
Answer: a) Encryption keys

134. Digital evidence admissibility depends on:

a) Proper handling
b) Fast processing
c) Printer logs
d) Cable replacement
Answer: a) Proper handling

135. Incident response planning improves:

a) Cyber resilience
b) Printer output
c) Cable speed
d) Storage size
Answer: a) Cyber resilience

136. Ethical hacking is different from malicious hacking because it is:

a) Legal
b) Anonymous
c) Hidden
d) Unauthorized
Answer: a) Legal

137. Reconnaissance can be:

a) Active
b) Passive
c) Both
d) None
Answer: c) Both

138. Network segmentation limits:

a) Attack spread
b) Printer speed
c) Cable length
d) OS performance
Answer: a) Attack spread

139. Security audits help:

a) Identify risks
b) Delete logs
c) Print reports only
d) Format drives
Answer: a) Identify risks

140. Phishing indicators include:

a) Suspicious links
b) Urgent requests
c) Unknown sender domains
d) All of the above
Answer: d) All of the above

141. Malware analysis is part of:

a) Digital forensics
b) Printing
c) Cable testing
d) OS installation
Answer: a) Digital forensics

142. Evidence examination should be done on:

a) Forensic copy
b) Original device
c) Printer
d) Public system
Answer: a) Forensic copy

143. Deepfake forensic detection checks:

a) Frame inconsistencies
b) Printer logs
c) Cable damage
d) Router speed
Answer: a) Frame inconsistencies

144. IoT botnets are commonly used for:

a) DDoS attacks
b) Printing
c) Cable theft
d) OS downgrade
Answer: a) DDoS attacks

145. OS downgrade attacks exploit:

a) Old vulnerabilities
b) New patches
c) Printer updates
d) Cable length
Answer: a) Old vulnerabilities

146. Cyber law ensures:

a) Legal regulation of cyberspace
b) Printer repair
c) Cable installation
d) OS deletion
Answer: a) Legal regulation of cyberspace

147. National Cyber Security Policy 2013 strengthens:

a) Cyber defense framework
b) Printer network
c) Cable quality
d) Router size
Answer: a) Cyber defense framework

148. Data breach penalties under DPDP can be:

a) Financial fines
b) Printer ban
c) Cable removal
d) OS suspension
Answer: a) Financial fines

149. Cybercrime against women includes:

a) Cyber stalking
b) Online harassment
c) Identity misuse
d) All of the above
Answer: d) All of the above

150. The main goal of ethical hacking is to:

a) Strengthen cybersecurity
b) Destroy systems
c) Steal data
d) Crash networks
Answer: a) Strengthen cybersecurity

151. Digital forensics supports law enforcement by:

a) Recovering digital evidence
b) Printing reports
c) Deleting logs
d) Formatting drives
Answer: a) Recovering digital evidence

152. Integrity of evidence is verified using:

a) Hash comparison
b) Printer logs
c) Cable ID
d) OS update
Answer: a) Hash comparison

153. A forensic workstation is used to:

a) Analyze copied evidence
b) Modify original data
c) Print documents
d) Delete files
Answer: a) Analyze copied evidence

154. Evidence contamination occurs when:

a) Original data is altered
b) Printer jams
c) Cable disconnects
d) Router fails
Answer: a) Original data is altered

155. Forensic timeline helps determine:

a) Attack sequence
b) Printer speed
c) Cable path
d) OS version
Answer: a) Attack sequence

156. A memory dump contains:

a) Active system processes
b) Printed logs
c) Cable data
d) BIOS firmware
Answer: a) Active system processes

157. Disk imaging tools must maintain:

a) Write protection
b) Data deletion
c) Printer control
d) Cable format
Answer: a) Write protection

158. Metadata can reveal:

a) File ownership
b) Creation date
c) Modification history
d) All of the above
Answer: d) All of the above

159. In ransomware investigation, first action is:

a) Isolate infected system
b) Pay ransom
c) Delete files
d) Restart system
Answer: a) Isolate infected system

160. Digital forensics ensures evidence is:

a) Court admissible
b) Random
c) Unverified
d) Deleted
Answer: a) Court admissible

161. Ethical hacking requires defining:

a) Scope of testing
b) Printer model
c) Cable size
d) OS brand
Answer: a) Scope of testing

162. Passive reconnaissance involves:

a) Public information gathering
b) Direct system attack
c) File deletion
d) Printer testing
Answer: a) Public information gathering

163. Active reconnaissance involves:

a) Direct interaction with target
b) Reading newspapers
c) Printing logs
d) Cable inspection
Answer: a) Direct interaction with target

164. Port scanning identifies:

a) Open network ports
b) Printer ink levels
c) Cable faults
d) Storage errors
Answer: a) Open network ports

165. Exploitation phase tests:

a) Vulnerability impact
b) Printer performance
c) Cable resistance
d) Router height
Answer: a) Vulnerability impact

166. Privilege escalation attack aims to:

a) Gain higher system access
b) Print documents
c) Delete logs
d) Restart OS
Answer: a) Gain higher system access

167. Maintaining access helps attacker test:

a) Persistence
b) Printer stability
c) Cable durability
d) Storage capacity
Answer: a) Persistence

168. Reporting phase should include:

a) Risk level
b) Impact analysis
c) Remediation steps
d) All of the above
Answer: d) All of the above

169. Risk assessment helps identify:

a) Potential threats
b) Printer faults
c) Cable damage
d) Router speed
Answer: a) Potential threats

170. Firewall primarily protects:

a) Network perimeter
b) Printer
c) Cable
d) Mouse
Answer: a) Network perimeter

171. AI phishing scams are dangerous because they are:

a) Personalized
b) Grammatically accurate
c) Context-aware
d) All of the above
Answer: d) All of the above

172. Ransomware encryption uses:

a) Cryptographic algorithms
b) Printer commands
c) Cable signals
d) Router settings
Answer: a) Cryptographic algorithms

173. DDoS attack aims to:

a) Overload server
b) Print files
c) Delete logs
d) Upgrade OS
Answer: a) Overload server

174. IoT botnets consist of:

a) Compromised smart devices
b) Printers only
c) Routers only
d) Servers only
Answer: a) Compromised smart devices

175. Deepfake fraud may involve:

a) CEO impersonation
b) Printer hacking
c) Cable theft
d) OS update
Answer: a) CEO impersonation

176. Secure firmware updates require:

a) Digital signatures
b) Printer logs
c) Cable marking
d) OS removal
Answer: a) Digital signatures

177. BIOS malware is dangerous because it:

a) Survives OS reinstall
b) Prints logs
c) Deletes files
d) Corrupts printer
Answer: a) Survives OS reinstall

178. WAF protects against:

a) Web application attacks
b) Printer hacking
c) Cable misuse
d) Router errors
Answer: a) Web application attacks

179. SQL injection targets:

a) Database queries
b) Printer spool
c) Cable network
d) OS boot
Answer: a) Database queries

180. Zero-day attacks are hard to prevent because:

a) No patch available
b) Printer error
c) Cable cut
d) Router crash
Answer: a) No patch available

181. NCSP 2013 promotes:

a) Public-private partnership
b) Printer networks
c) Cable design
d) OS printing
Answer: a) Public-private partnership

182. Critical Information Infrastructure includes:

a) Banking systems
b) Power grids
c) Defense networks
d) All of the above
Answer: d) All of the above

183. IT Act 2000 supports:

a) E-commerce
b) E-governance
c) Digital signatures
d) All of the above
Answer: d) All of the above

184. Section 43 penalizes:

a) Unauthorized access
b) Printer damage
c) Cable theft
d) OS deletion
Answer: a) Unauthorized access

185. Section 69 authorizes:

a) Data interception
b) Printer shutdown
c) Cable removal
d) Router deletion
Answer: a) Data interception

186. DPDP Act emphasizes:

a) Data privacy rights
b) Printer control
c) Cable inspection
d) OS updates
Answer: a) Data privacy rights

187. Consent under DPDP must be:

a) Clear and informed
b) Hidden
c) Implied only
d) Verbal only
Answer: a) Clear and informed

188. CCPWC scheme focuses on:

a) Cyber safety awareness
b) Printer usage
c) Cable repair
d) Router upgrade
Answer: a) Cyber safety awareness

189. Cyber harassment includes:

a) Online abuse
b) Cyber stalking
c) Threat messages
d) All of the above
Answer: d) All of the above

190. Cyber terrorism under Section 66F includes:

a) Threat to national security
b) Printer hacking
c) Cable theft
d) OS downgrade
Answer: a) Threat to national security

191. Forensic documentation must be:

a) Accurate
b) Detailed
c) Time-stamped
d) All of the above
Answer: d) All of the above

192. Evidence preservation ensures:

a) No alteration
b) Fast printing
c) Cable fixing
d) OS update
Answer: a) No alteration

193. Digital investigation models provide:

a) Structured approach
b) Printer workflow
c) Cable mapping
d) OS setup
Answer: a) Structured approach

194. Forensic readiness reduces:

a) Investigation delay
b) Printer ink
c) Cable usage
d) Router size
Answer: a) Investigation delay

195. Ethical hacking certifications include:

a) CEH
b) OSCP
c) CISSP
d) All of the above
Answer: d) All of the above

196. Social engineering prevention requires:

a) Awareness training
b) Printer upgrade
c) Cable testing
d) OS removal
Answer: a) Awareness training

197. Cyber resilience means:

a) Ability to recover quickly
b) Printer repair
c) Cable replacement
d) OS downgrade
Answer: a) Ability to recover quickly

198. Incident response plan includes:

a) Detection
b) Containment
c) Recovery
d) All of the above
Answer: d) All of the above

199. Digital forensic investigation ends with:

a) Case closure
b) Printer shutdown
c) Cable removal
d) OS update
Answer: a) Case closure

200. The ultimate objective of UNIT 5 concepts is:

a) Secure and legally compliant cyberspace
b) Printer maintenance
c) Cable control
d) OS deletion
Answer: a) Secure and legally compliant cyberspace